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A Protein First Computational Screen for Structured RNA Elements

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Abstract

RNA underpins many of the most transformative advances in modern biotechnology, including RNA interference, CRISPR systems, metabolite-sensing riboswitches, and recently discovered Bridge RNA-guided recombination systems. Despite their importance, discovering new functional RNAs remains challenging. Unlike proteins, RNA sequences often evolve rapidly and are conserved primarily through structure rather than sequence, limiting the effectiveness of conventional homology-search methods. In contrast, protein homology detection is a more tractable problem with robust and fast implementations.

Here, we present a protein-first RNA discovery framework that leverages conserved bacterial protein domains to identify candidate structured RNAs within neighbouring non-coding regions. We use comparative genomics and identify covarying sites, which mutate together across evolution to preserve base pairing, as evidence of real structure. As an initial application, we focus on light-associated protein systems to search for novel light-responsive RNA regulators or RNA-protein complexes, in effect looking for potential RNA light switches. Two weak BLUF-associated candidates were identified, and the FMN riboswitch positive control was rediscovered, while the negative controls yeilded no strong hits. We further introduce an approach for measuring relatedness between RNA elements, addressing a gap in this area. Together, this framework offers a targeted and potentially scalable route to RNA discovery, allowing biological pathways of interest to be screened for previously uncharacterized functional RNA elements.

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Abbreviations

MSA	Multiple Sequence Alignment	20
HMM	Hidden Markov Model	16
CM	Covariance Model	12
GTDB	Genome Taxonomy Database	11
BLAST	Basic Local Alignment Search Tool	5
NCBI	National Center for Biotechnology Information	12
COVID-19	Coronavirus Disease 2019	2
CRISPR	Clustered Regularly Interspaced Short Palindromic Repeats	viii
crRNA	CRISPR RNA	2
DMS-MaPseq	Dimethyl Sulfate Mutational Profiling with Sequencing	7
HDV	Hepatitis Delta Virus	1
mRNA	Messenger RNA	2
RNAi	RNA Interference	2
RNP	Ribonucleoprotein	v
rRNA	Ribosomal RNA	2
SELEX	Systematic Evolution of Ligands by Exponential Enrichment	7
SHAPE-seq	Selective 2'-Hydroxyl Acylation analysed by Primer Extension Sequencing	7
sRNA	Small Regulatory RNA	2
KS	Kolmogorov-Smirnov	33
BLUF	Blue-Light Using FAD	v
FAD	Flavin Adenine Dinucleotide	iv
iTOL	interactive Tree Of Life	22
FMN	Flavin Mononucleotide	v
SAM	S-adenosylmethionine	18
GB	Gigabyte	26
MB	Megabyte	26
CPU	Central Processing Unit	26

ETE Environment for Tree Exploration 17
ipTM interaction predicted Template Modelling score 26
PCR Polymerase Chain Reaction 61
AUC Area Under the Curve 33

1. Introduction

RNAs play a wide range of roles in cellular processes, many of which have been harnessed for biotechnological applications, and are likely to have many more functions yet to be discovered. This project aims to identify novel conserved RNA structural elements associated with light-sensing gene families in bacteria. This introduction provides background and rationale for the screening of novel structured RNA elements in bacterial genomes, and outlines the aims and significance of the project.

1.1 Ribonucleic acids

The central dogma of molecular biology, first articulated by Francis Crick, describes the flow of genetic information: DNA $\rightarrow$ RNA $\rightarrow$ Protein[1]. In this classical view, RNA serves primarily as an intermediate messenger, transcribing genetic information from DNA and translating it into proteins at the ribosome. However, this perspective underrepresents the diversity in the roles of RNA molecules.

In the hypothetical RNA world, RNA molecules are thought to have performed many of the roles now largely carried out by proteins and DNA, including catalysis, regulation, and information storage[2–5]. Although these functions have been largely superseded by proteins and DNA, numerous RNA-based regulatory mechanisms persist in modern organisms.

1.1.1 Functional RNAs in biology and biotechnology

The diverse functional roles of RNA are driven by their structural and reactive diversity. Naturally occurring functional RNAs include purely RNA systems, riboswitches and ribozymes, and those in complexes with proteins, while *in vitro* selected RNAs include fluorogens and aptazymes[6].

Ribozymes are catalytic RNAs that accelerate chemical reactions, much like protein enzymes. Beyond the ribosome and spliceosome, several small self-cleaving ribozymes exist in nature, including the hammerhead, hairpin, Hepatitis Delta Virus (HDV), glmS, twister, and pistol ribozymes[6–8].

RNA has been central to several major biotechnological breakthroughs. CRISPR-

Cas9 gene editing harnesses guide RNAs for programmable genome modification[9]. Messenger RNA (mRNA) vaccines against Coronavirus Disease 2019 (COVID-19) demonstrated the therapeutic potential of synthetic mRNAs[10]. RNA Interference (RNAi) enables targeted gene silencing in research and therapeutics[11].

RNA aptamers such as Spinach bind fluorogenic compounds and fluoresce upon binding, enabling RNA visualization in living cells[12]. The photoriboswitch Werewolf-1, generated through *in vitro* selection, responds to light by binding photoisomerizable ligands, offering potential for optogenetic control of gene expression[13]. These RNAs, involved in fluorescence and photoreception, have broad relevance for medical research, biotechnology, and agricultural applications[6].

1.1.2 Ribonucleoprotein complexes

While some RNAs function independently, many operate in RNP complexes. The ribosome is one such RNP with critical function in all cells. This large complex catalyses peptide bond formation, with Ribosomal RNA (rRNA) forming the catalytic core, while ribosomal proteins provide structural support and regulatory functionality[14]. Spliceosomes are another example of essential RNPs in eukaryotes[14].

In bacteria, several classes of RNPs mediate post-transcriptional regulation. Small Regulatory RNAs (sRNAs) often require the RNA chaperone Hfq to facilitate base pairing with target mRNAs, thereby modulating transcript stability[15].

CRISPR-Cas systems exemplify the complexity of bacterial RNPs. CRISPR RNAs (crRNAs) guide Cas nucleases to complementary DNA or RNA targets, enabling sequence-specific nucleic acid cleavage[16]. This mechanism has famously been repurposed as a gene-editing tool[9].

Durrant et al.[17] recently discovered Bridge RNA, which forms a complex with a transposase. This system was found in a family of minimal insertion sequences, IS110, which expresses a transposase that is essential for the mobility of the insertion sequence. Prior to experimental validation, the discovery was done by a computational screen for transposase homology and subsequent computational identification of conserved RNA structured elements directly flanking the transposase genes. Bridge RNA systems are further shown to be reprogrammable to offer targeted control of transposase activity, enabling precise genome editing through site-specific insertion, excision and inversion. This is a major finding in the field of gene editing, adding to the library of tools available for precise genome manipulation.

1.1.3 Cis-regulatory RNAs

Cis-regulatory RNAs are structured RNA elements that regulate the expression of genes located on the same transcript. Riboswitches are a well-studied class of cis-regulatory RNAs that modulate gene expression in response to some stimuli, often metabolites[18]. Although seemingly less abundant than protein factors that perform similar functions, riboswitches are recognized as important regulatory elements across all domains of life. They are typically embedded within messenger RNAs and control gene expression through ligand-induced conformational changes.

Transcriptional riboswitches control formation of Rho-independent terminators — intrinsic termination structures consisting of a hairpin followed by a uridine-rich tract. Depending on the riboswitch, ligand binding may promote terminator formation or stabilize an antiterminator structure, enabling transcriptional read-through. Similarly, translational riboswitches can sequester or expose the ribosome-binding site in response to ligand binding[18].

1.1.4 RNA structure determines function

RNA molecules, much like proteins, fold into complex structures that enable diverse functions[19]. They typically exist as single-stranded molecules, which can fold back on themselves through intramolecular base pairing to form secondary and tertiary structures. RNA secondary structure arises from Watson-Crick base pairing (A-U and G-C) and wobble pairing between nucleotides within the same strand[20]. Figure 1.1 illustrates the diversity of RNA secondary structure elements, which enable a variety of functions, enabling interactions with proteins, other RNAs, and small molecules[21]. Common secondary structures include hairpins, consisting of a base-paired stem capped by an unpaired loop; internal loops and bulges, made up of unpaired nucleotides within a helical stem; junctions where three or more helical stems converge; and pseudoknots, where nucleotides in a loop pair with regions outside that loop[22].

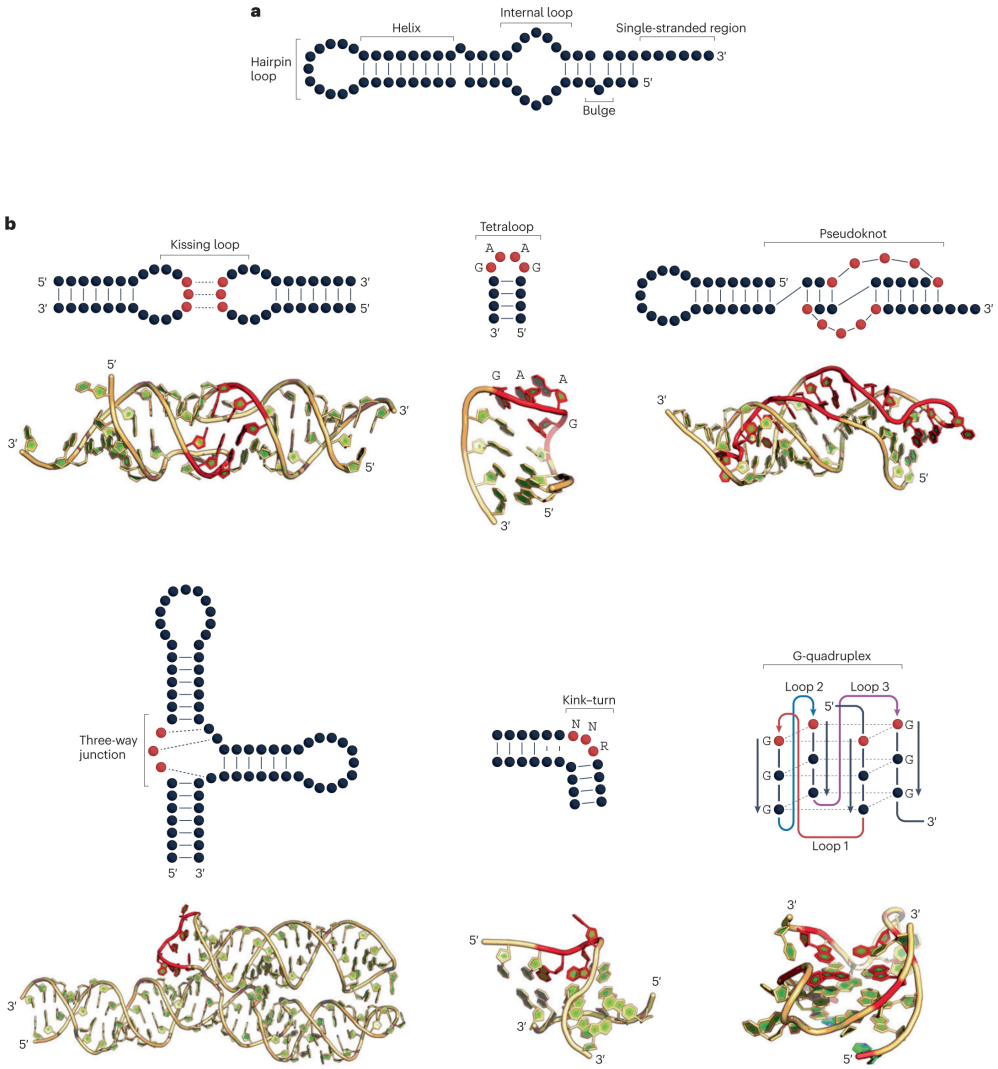


Figure 1.1: The diversity of RNA secondary structure elements. RNA molecules fold into complex structures through intramolecular base pairing, forming secondary structures such as hairpins, internal loops, bulges, junctions, and pseudoknots. **a)**, Common secondary structure elements include hairpins, internal loops, bulges, and single-stranded regions. **b)**, Examples of more complex secondary structure elements: kissing loops, tetraloops, pseudoknots, junctions, kink turns and G-quadruplexes. Also shown are 3D structures of specific examples of these motifs from the Protein Data Bank. Red nucleotides indicate the distinct structural motifs. Modified from Marklund et al. [21].

Beyond secondary structure, RNAs fold into more complex tertiary structures through long range interactions [23]. These include base triplets, ribose zippers, A-minor motifs, and tetraloop-receptor interactions, which stabilize compact RNA folds.

Although there are many examples of structured functional RNAs, it is important to note there are many cases where sequence alone provides function [11;24]. The relationship between RNA structure and function is exemplified by riboswitches, a class of regulatory RNAs. The aptamer domain folds into a specific three-dimensional pocket

that selectively binds a target ligand through a combination of electrostatic interactions and shape complementarity[25]. Ligand binding stabilizes one conformational state over another, thereby modulating transcription termination or initiation. This allosteric mechanism, whereby binding at an aptamer site affects a distal functional region, demonstrates how RNA structure enables regulatory function.

1.2 Current approaches to functional RNA discovery

It is predicted that a large number of functional RNAs remain undiscovered, potentially on the order of thousands[25]. Breaker and colleagues have pioneered systematic riboswitch and ribozyme discovery, identifying over 55 distinct riboswitch classes as of 2022[26]. Other major RNA-related discoveries, such as ribosomes, CRISPR-Cas systems, and Bridge RNA, also occurred through exploratory research approaches[27].

1.2.1 Computational discovery approaches

Bioinformatic sequence-based searches using tools like Basic Local Alignment Search Tool (BLAST) identify homologous sequences, but many functional RNAs have divergent sequences detectable only through structural conservation. Structure based searches have proven successful, like the bioinformatic search for twister ribozymes by Roth et al.[7] discovering roughly 2,700 ribozyme candidates. These techniques search on sequence and secondary structure patterns, but require existing alignments and structure models, limiting their utility for discovering entirely novel RNA classes.

RNA structure prediction remains a challenging problem, particularly when given a single sequence and no homology structures are known, as would be the case with a genome wide search for novel RNA structures in a single species[24]. Comparative approaches using multiple homologous sequences provide greater predictive power for RNA structure inference[28]. Figure 1.2 illustrates the idea of covariation and how it can be used to identify conserved RNA structures across related species. Where pairs of nucleotides covary across species, i.e., they mutate together to preserve base pairing, this provides evidence of conserved RNA structure.

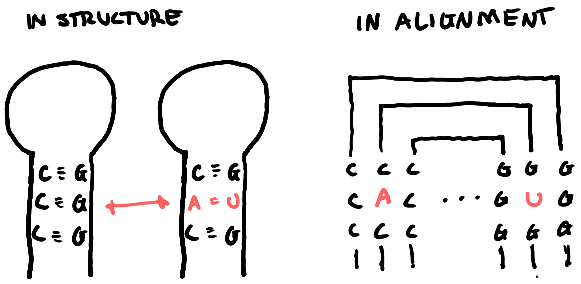


Figure 1.2: Identification of covarying sites in RNA alignments. Covarying sites are pairs of nucleotides that mutate together across evolution to preserve base pairing, providing evidence of conserved RNA structure. In this example, a C-G base pair in one species is replaced by an A-U base pair in another species, maintaining the overall secondary structure of the RNA. Covariation analysis is a powerful tool for identifying conserved RNA structures across related species, enabling the discovery of conserved structural features.

The CMFinder tool Yao et al.[29], investigates covariance across multiple sequences, enabling the identification of conserved structural features. These conserved regions show evidence of selective constraint on structure, indicating potential functional importance. Such RNA elements may represent riboswitches, ribozymes, or other classes of functional RNAs. The Weinberg lab has had several notable successes with their tool, CMfinder,[7;8;30;30]. Roth et al.[7] also employed CMfinder to identify a new class of self cleaving ribozymes, the twister ribozyme, which has since been found to be widespread in nature. In that paper hammerhead and twister ribozymes were observed to commonly reside near each other and to certain protein coding genes, although noting it still a biological mystery. Weinberg et al.[8] followed up on this observation with a screening of sequences nearby known ribozyme and ribozyme associated protein domain coding regions. This approach successfully identified additional ribozyme classes including pistol and hatchet. The 2025 paper, Sack et al.[31], recently identified 16 new structured RNA classes in plants using a CMfinder-based bioinformatic screen. This study focused on structured RNAs involved in alternative splicing.

1.2.2 Experimental approaches to discovery and generation of functional RNAs

Transcriptome mapping approaches have proven successful for unbiased riboswitch discovery. Term-seq maps RNA 3' termini genome-wide to identify premature transcription termination sites characteristic of riboswitches and attenuators[32]. This approach has uncovered numerous antibiotic-responsive riboswitches controlling resistance genes in bacterial pathogens and within the human microbiome. RNA struc-

ture probing methods characterize secondary structures, tertiary interactions, and dynamic conformational changes. Selective 2'-Hydroxyl Acylation analysed by Primer Extension Sequencing (SHAPE-seq) chemically modifies unpaired nucleotides, enabling genome-wide structure mapping[33]. This approach can identify structured regions throughout transcriptomes, revealing RNA switches, ribozymes, and protein-binding sites. Dimethyl Sulfate Mutational Profiling with Sequencing (DMS-MaPseq) extends this approach to structural probing *in vivo*[34]. A variety of methods exist for studying protein-RNA interactions from both protein- and RNA-centric approaches[35]. While powerful for validation and mechanistic characterization, computational approaches have been successful in extending the scope of RNA research[24].

Systematic Evolution of Ligands by Exponential Enrichment (SELEX) is a technique to generate novel functional RNAs by applying selective pressure to a library of random sequences. This technique was used in the generation of the photoriboswitch, Werewolf-1[13]. Generation of synthetic functional RNAs by computational design has also been reported, with tools like RfamGen and RhoDesign[36].

Many RNA discoveries, including CRISPR-Cas systems, mRNA vaccine strategies and many catalytic RNAs were however discovered in nature, not designed *de novo*[6;16]. Beyond biotechnology, fundamental understanding of natural RNA-based systems has value, for example the discovery of flouride sensing riboswitches in bacteria

1.3 The Twilight Zone of RNA and protein homology

Homology is being used here as a catch all term for sequence and structural similarity due to shared ancestry. Orthology, parology and xenology are all included in this broad definition of homology. Othologs are are genes which have diverged through speciation events[37]. Conversly, paralogs are genes that have diverged through duplication events within a species[37]. Xenology is a term used to describe genes that have been transferred between species through horizontal gene transfer[37]. Homoloy is often infered from sequences or structures share more similarity than would be expected by chance[38]. Homolgous genes have shared ancestry and potentially shared function and evolutionary pressure, but the processes that have led to their divergence can make it difficult to detect homology through sequence similarity alone.

The Twilight Zone of homology detection refers to the range of sequence similarity where it is difficult to confidently infer homology based on sequence alone[39]. The Twilight Zone of protein homology was estimated by Rost[39] at 20-35% sequence identity, where the probability of correctly identifying homologs drops significantly. This means that even if two homologous proteins can be identified as such until they have diverged to the point where their sequences are less than 20-35% identical.

RNA homology detection is one way of identifying functional RNAs, as conservation across homology can detect similar selective pressures and functionality. Further, with homologous sequences found across multiple species, comparative approaches can be used to identify conserved RNA structures[24]. RNA homology searches are complicated by the fact that RNA structure is often more conserved than sequence . The Twilight Zone of RNA homology detection is less well defined than for proteins, but was estimated by McGimpsey[40] to be around 35-50% sequence identity. This means it is more difficult to detect deep homology in RNA sequences than in protein sequences.

1.4 A protein-first approach to RNA discovery

The approach being proposed here is a protein-first approach to RNA discovery, leveraging the relative ease of protein homology detection to identify conserved protein domains and deeper evolutionary relationships, and then searching for conserved RNA structures in neighbouring non-coding regions (Figure 1.3). Beyond the successes of gene family-targeted searches with CMfinder, there is precedent for riboswitches being encoded upstream of certain gene families[6;25]. This approach also provides a less computationally intensive alternative to genome-wide screens, allowing for wider searches across more species. It harnesses the power of evolutionary conservation and leverages the biological assumption that functionally related genes (i.e. with domain homology) may share regulatory mechanisms.

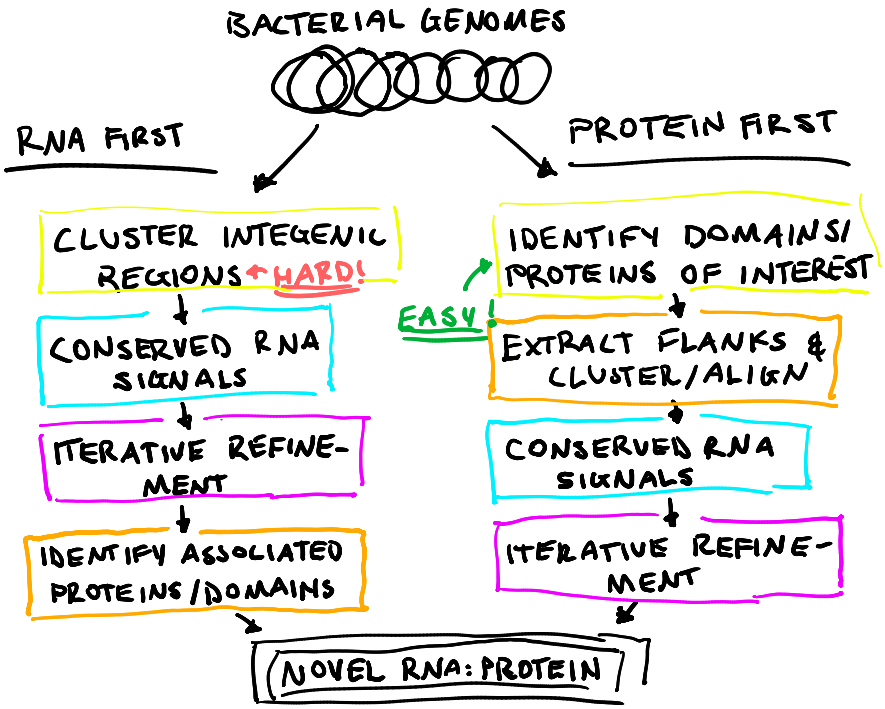


Figure 1.3: A protein-first approach to RNA discovery. Typical RNA discovery approaches are RNA-first, searching for conserved RNA structures across genomes and then associating them with nearby genes. In contrast, a protein-first approach leverages the relative ease of protein homology detection to identify conserved protein domains and deeper evolutionary relationships and then searches for conserved RNA structures in neighbouring non-coding regions.

Because of the potential importance of light-sensing systems in biotechnology, this project screened for conserved RNA structures located near light-sensing gene families. Light-sensing systems in bacteria often involve complex regulatory networks, and RNA-based mechanisms have been shown to play roles in light acclimation responses in cyanobacteria[41].

A recently published study by a group in Copenhagen, Geissler et al.[42], describes a bioinformatic screen similar to the one undertaken here. In that work 931 orthologous cyanobacteria genes involved in various pathways were targeted for nearby conserved RNA structure discovery using CMfinder. Approximately 400 novel conserved RNA structures were identified, some of which were in intergenic regions up and downstream of light-sensing and photosynthetic genes. This work demonstrates the potential for searching for conserved RNA structures near light-sensing gene families, and the potential for discovering novel regulatory elements in these systems. The approach taken in that study was, however, more of an RNA-first discovery pipeline where large sets of non-coding regions from a relatively small number of genomes (202) were screened for conserved RNA structures. Filtering for GC content and other

non-structural features was used to reduce the number of candidate structures.

With a protein-first approach, the search spaces can be expanded to include a much larger number of genomes, and the search can be targeted to specific protein domains of interest. With a larger search space and a more targeted approach, the power of a covariance analysis is expected to increase, as power depends on the number and phylogenetic diversity of the sequences analysed. More diverse sequences give more opportunities to observe covarying positions, which should improve the statistical significance of the results and the chance of detecting biologically relevant signals.

1.4.1 A search target: Blue Light sensor Using FAD

The BLUF domain is a light-sensing protein domain that binds flavin adenine dinucleotide (FAD) and undergoes a conformational change upon blue light absorption[43;44]. This conformational change can modulate the activity of associated effector domains, enabling light-dependent regulation. BLUF was chosen as a proof-of-concept search target for its potential to be involved in an RNP 'light switch' complex.

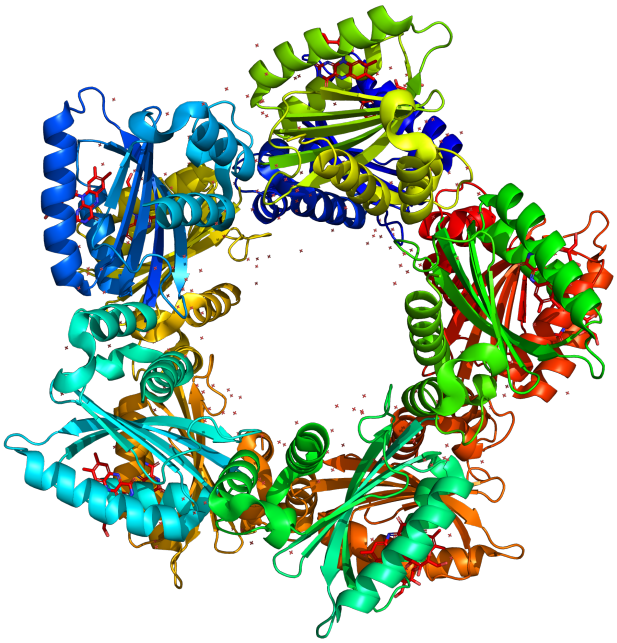


Figure 1.4: BLUF structure. Crystal structure of the BLUF domain from *Thermosynechococcus elongatus* BP-1[43]. Ten Tll0078 monomers with N-terminal BLUF domains and C-terminal domains form two pentameric rings.

Figure 1.4 shows the crystal structure of the BLUF domain from *Thermosynechococcus elongatus* BP-1, highlighting its characteristic fold. The domain is composed of a five-stranded β-sheet flanked by two α-helices, forming a compact structure that binds FAD. Three conserved residues, Asn31, Asn32 and Gln50, are thought to be

critical for flavin binding and light-induced conformational changes[43].

1.4.2 Bacterial genomes as a search space

Riboswitches have been most extensively discovered and experimentally validated in bacteria and seem to rarely have homologs in eukaryotes[25]. This makes bacteria a reasonable domain for riboswitch discovery efforts.

Beyond riboswitches, many other functional RNAs in bacteria act through protein partners. Small RNAs and RNA-binding proteins such as Hfq and CsrA form extensive post-transcriptional regulatory networks[15;45;46], and the RNA component of ribonuclease P is a catalytic RNA-protein complex conserved across bacteria[3]. CRISPR-Cas systems, in which guide RNAs direct protein effectors to their targets, are also widespread in bacterial genomes[9;16], and new RNA-guided systems continue to be described[27]. This provides rationale for searching for conserved RNA structured elements in bacterial genomes.

Extensive phylogenetic diversity is available among sequenced bacterial genomes. As of April 2026, over 870,000 bacterial genomes spanning 180,000 bacterial species clusters have been sequenced and organised in databases such as Genome Taxonomy Database (GTDB)[47]. This represents vast evolutionary distances, bacteria diverged over 3 billion years ago, providing the deep phylogenies necessary for detecting conserved structures through covariation analysis.

Though bacterial regulation is not simple[24;48;49], unlike eukaryotes, bacterial mRNAs typically lack capping, splicing, and nuclear export[14;50], and transcription and translation are often coupled[51;52]. This may facilitate the identification of conserved RNA elements from genomic DNA sequences, as fewer processing steps (e.g. splicing) complicate the relationship between genomic sequence and mature transcript.

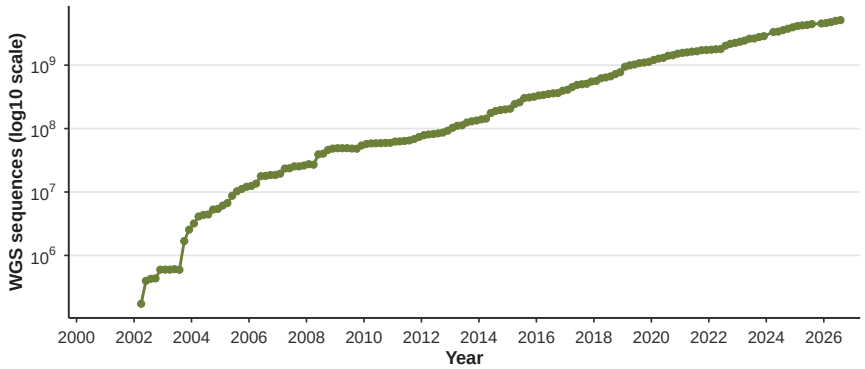


Figure 1.5: The number of bacterial whole genomes sequenced since 2000. The number of bacterial whole genomes available from GenBank releases over time. WGS = whole genome sequencing, shown on a $\log_{10}$ scale. Graphed from data publicly available from https://www.ncbi.nlm.nih.gov/genbank/statistics/.

Figure 1.5 illustrates the rapid increase in the number of bacterial genomes sequenced over the past two decades and available on National Center for Biotechnology Information (NCBI)'s GenBank[53], highlighting the growing availability of genomic data for comparative analyses. The set of available bacterial genomes for screening is constantly expanding, providing an ever-increasing resource for the discovery of conserved RNA elements.

1.4.3 Bioinformatic approach

The right hand path of Figure 1.3 illustrates the protein-first approach to RNA discovery taken in this project. The general approach is to identify conserved protein domains, and then search for conserved RNA structures in neighbouring non-coding regions. The first step is to identify conserved protein domains of interest, such as BLUF, across a wide range of bacterial genomes. Tools such as FASTA and BLAST can be used to identify homologous proteins by comparing sequences, while HMMER uses both sequence and structure information from probabilistic models to identify homologous protein domains[38]. Durrant et al.[17] had success using HMMER in their protein-first approach, and this approach will be used here as well. Once these domains are identified, the genomic regions flanking these genes can be extracted and analyzed for conserved RNA structures using comparative genomics approaches. These include tools such as CMfinder, which identifies conserved RNA secondary structures using a comparative genomics approach[29]. Alternatively, clustering and alignment tools (MAFTT and MMseqs2[54;55]) can be used to identify local sequence homology before searching for conserved RNA structures in these regions by covariance analysis. Several tools exist for covariance analysis, including R-scape, which uses statistical models to identify covarying base pairs (Figure 1.2) in RNA alignments[56]. This second apprach was used in this project after initial attempts to use CMfinder were unsuccessful, and is described in more detail in the methods section.

Probabilistic models of RNA structure and sequence can be built from identified structural element candidates, most commonly these are Covariance Model (CM)s. CMs are statistical models that capture both the sequence and structural features of RNA families, enabling sensitive searches for homologous sequences in genomic databases[57].

1.5 Project aims and hypothesis

The overarching aim is to discover novel structured RNA elements associated with bacterial light-sensing systems by leveraging comparative genomics. Gene-family-targeted approaches offer a computationally tractable approach for identifying regulatory elements linked to specific biological pathways. Light-sensing gene families are of

particular interest due to their importance in environmental adaptation and their potential relevance to biotechnological applications. Identifying conserved RNA structures associated with these genes may reveal previously uncharacterized regulatory mechanisms. The first aim of this project is therefore to screen for conserved RNA structural elements located near the BLUF domain across diverse bacterial genomes. By examining intergenic and transcript regions associated with these genes, the project aims to predict conserved RNA secondary structures using comparative genomics approaches that identify covariation patterns indicative of selective constraint. The second aim is to develop a protein-first structured RNA discovery pipeline, capable of systematically screening for conserved RNA structures. This pipeline is intended to be reusable and adaptable for future studies, targeting additional gene families or biological pathways. Together, these aims will contribute both new candidate structured RNA elements and a methodological platform for search for structured RNAs in nature.

Further, this work aims to fill a gap in the field for a reliable statistical method for measuring relatedness between RNA elements, which is important for understanding the evolutionary relationships and structural similarities among RNA motifs. This has remained a challenge in the field despite the development of several methods for RNA structured element comparison[58;59].

The central hypothesis of this work is that protein homology can be leveraged in a discovery pipeline for identifying RNA homology in RNP systems. This will be tested by recovery of a known structured RNA element. This relies on bioinformatic tools being able to identify conserved protein domains and then search for conserved RNA structures in neighbouring non-coding regions. Testing if any identified conserved RNA structures have any function or are involved in light sensing is beyond the scope of this project, but the discovery of novel conserved RNA structures will provide a foundation for future experimental validation and functional characterization.

1.6 Vision Mātauranga

This project uses publicly available genomic datasets. If genomes from Aotearoa-associated or taonga species are included, appropriate consideration will be taken regarding data sovereignty and potential social and cultural outcomes. A Ngāi Tahu Research Consultation Committee application is in progress regarding the larger, overarching work this project falls under.

The research in this project aligns most closely with the Vision Mātauranga theme of indigenous innovation, contributing computational tools with potential applications in biotechnology and plant science in Aotearoa New Zealand. We remain open to collaboration and consultation with tāngata whenua. Should commercial opportunities

arise as a result of this research, we will seek to ensure that benefits are shared equitably with iwi and other relevant stakeholders.

2. Methods

The following Methods chapter outlines: firstly the production of a positive-control set of known RNA elements and their associated protein domains. Secondly the discovery pipeline for identifying novel structured RNA elements associated with conserved protein domains, and thirdly a description of an iterative expansion approach to amplifying the identified candidate signal. The main computational discovery pipeline uses data curated from public datasets to screen for conserved structured RNA elements encoded near a conserved protein domain anchor. Finally, a novel approach to clustering RNA covariance models was developed and applied to the candidate set of structured RNA elements identified by the discovery pipeline. Candidate RNAs were also assessed for possible interaction with their anchor protein using AlphaFold 3 predictions. An overview of the methods used in this study is provided in Figure 2.1.

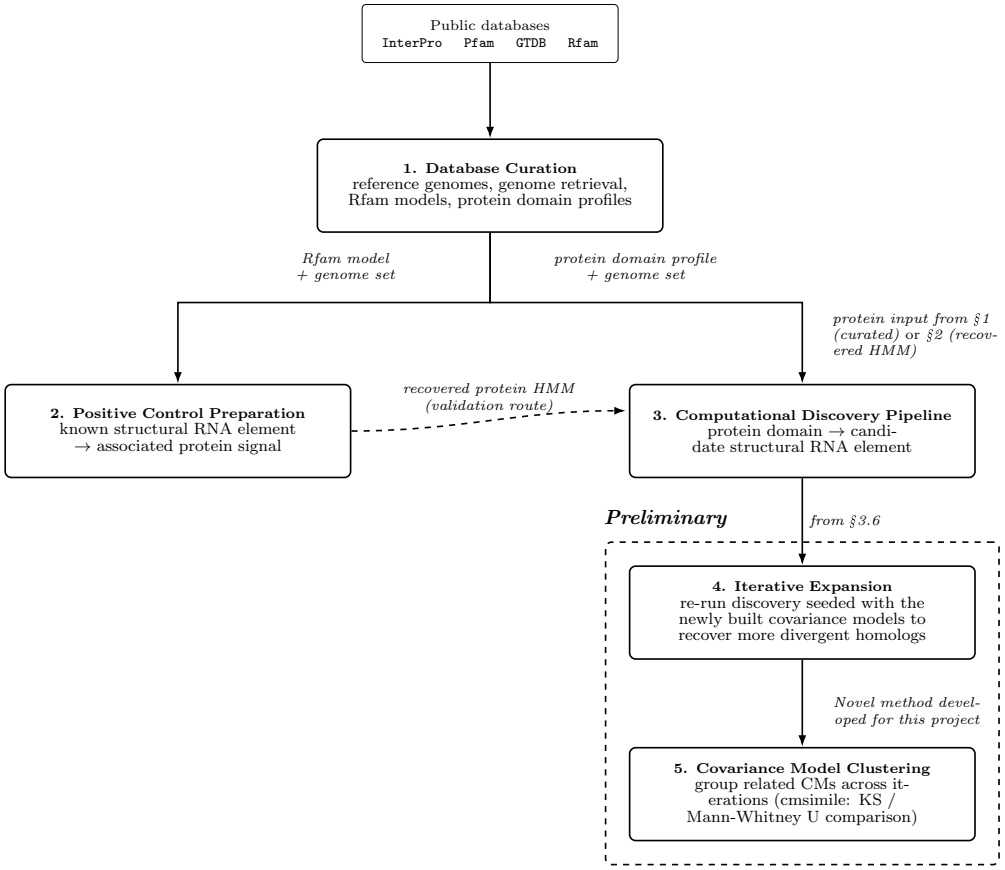


Figure 2.1: Overview of methods Simplified overview of the approach taken for RNP candidate discovery. Numbered steps refer to the following subsections. Public databases were used to curate reference genomes, protein domain Hidden Markov Model (HMM)s and RNA covariance models. A positive control pipeline was developed to validate the association between known cis-regulatory RNA elements and their associated protein domains. A computational discovery pipeline was developed to identify novel structured RNA elements associated with conserved protein domains. An iterative expansion approach was developed to amplify the signal of the identified candidates, and a novel clustering approach was developed to group related covariance models. Italic labels show the data passed between steps.

The discovery and positive control pipelines are written in Python 3.14.6 using Snakemake v.9.23.1, and are configurable and automated[60]. Iterative expansion and covariance model clustering were developed in the Go programming language v.1.27.1 and are available as standalone tools, both of which are highly configurable and reusable. All code, software environment definitions and data used in this study are available in the supplementary materials (Appendix 5.1).

2.1 Database curation

2.1.1 Reference genomes

For all analyses, the GTDB release 232[61] was used to select a diverse set of bacterial genomes for analysis. The GTDB provides a standardised taxonomy, ensuring that the selected genomes are representative of the bacterial domain. Python package Environment for Tree Exploration (ETE) v.3.1.3[62] was used for phylogenetic analysis and filtering using the GTDB phylogenetic tree. Further filtering was performed to select high-quality, complete genomes (completeness $> 99.9\%$, contamination $< 1\%$), ensuring the selected reference genomes are suitable for downstream analysis. Completeness and contamination metrics were provided by CheckM2[63], available in the GTDB release.

Only genomes with genomic sequence, protein sequence and annotation files available were retrieved from the NCBI GenBank database release 265.0[53]. Full lists of genomes used in this study for particular experiments are available in the supplementary materials (Appendix 5.1).

2.1.2 Reference protein domain profiles

Protein domain HMMs are probabilistic models that describe the conserved sequence of a family of protein domains. InterPro (version 35.0) entries guided the choice of protein domain model and relevant genome search space[64]. Protein domain HMM profiles were retrieved from the Pfam v.35.0 database[65]. These were used in the discovery pipeline to search for protein homology to anchor the search for conserved structured RNA elements.

2.1.3 Reference RNA covariance models

Covariance models are probabilistic models that describe the conserved sequence and secondary structure of a family of RNA elements. RNA family covariance models were retrieved from the Rfam database (version 15.0)[66]. These were used in the positive control pipeline to identify known RNA elements in the reference genome set, and to extract flanking coding sequences to identify associated protein domains. These were used in positive control experiments to validate the pipeline and to identify known RNA families among the candidates identified.

Additionally, covariance models and clan information available from the Rfam database (version 15.0)[66] were used to validate the clustering method developed in this study. Seed sequences (curated members of each family, used to build the alignments from which the curated consensus models are built) were also retrieved.

2.2 Positive control preparation

The positive control pipeline was developed to validate the association between known cis-regulatory RNA elements, the FMN, S-adenosylmethionine (SAM) and fluoride riboswitches, and CRISPR-Direct Repeat 42, to their associated protein domains. This was to validate the premise of this study, that conserved structured RNA elements are associated with conserved protein domains, and to provide a set of validated examples to test the discovery pipeline.

Pipeline inputs are Rfam covariance models and annotated seed sequences, as described in Section 2.1.3. A custom script accessing Entrez via the Biopython v.1.87 Python library searches for genomes containing these sequences[53]. A genome set is then constructed from GTDB reference genomes and downloaded from GenBank (Section 2.1.1). All inputs from runs of this pipeline used in this study are available in Appendix Table 5.2.

2.2.1 RNA structure search and extraction of flanking coding sequences

Members of the Rfam family were searched using the Rfam published search parameters, to revalidate hits and confirm the coordinate system in the reference genomes (Infernal v.1.1.5 *cmsearch*). The output of this step is a list of genomic coordinates for each hit, which is used to extract the flanking coding sequences.

Coding sequences surrounding the RNA element locus were extracted from the genome, within a 1000-nucleotide region of the element boundaries. This was done to identify protein domains that are encoded in close proximity to the RNA element, which may indicate a functional relationship between the two. The approach is illustrated in Figure 2.3.

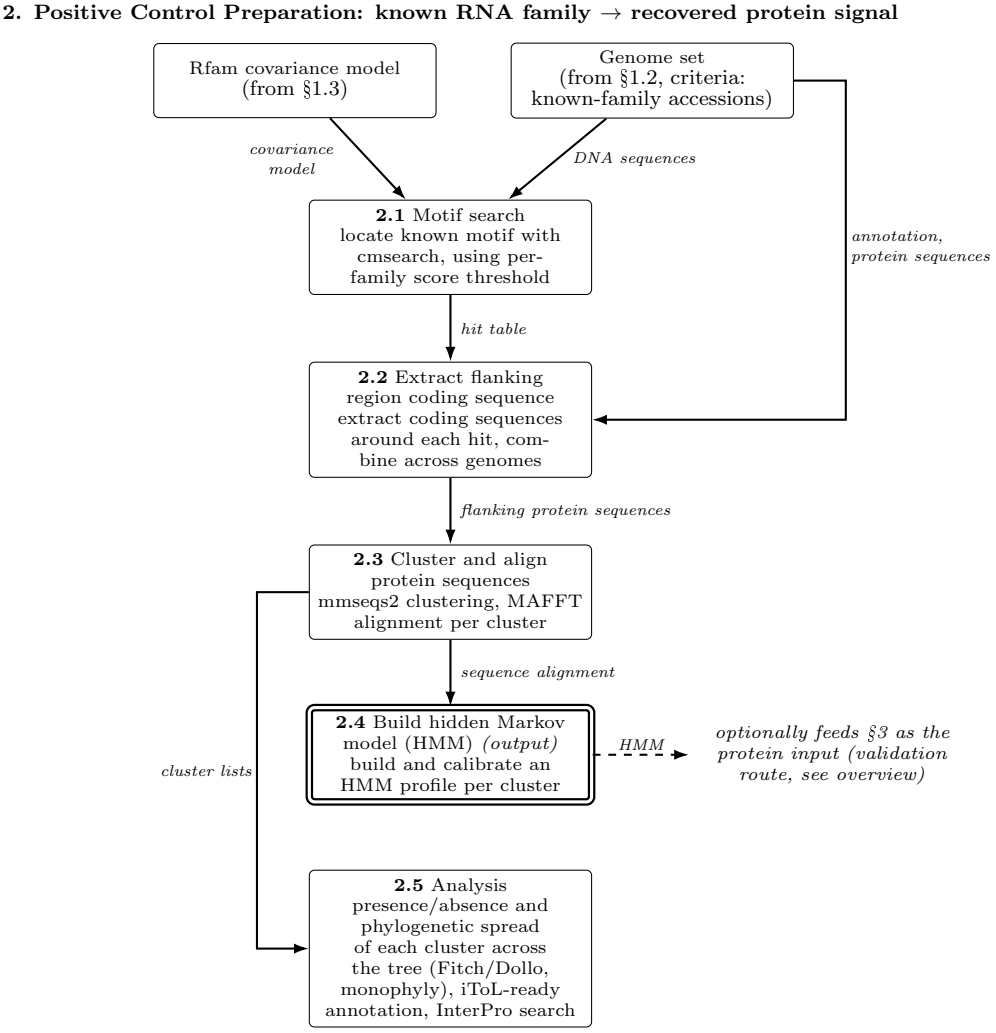


Figure 2.2: Positive control preparation (known RNA family → recovered protein signal). Starting from Rfam covariance models (§2.1.3) and genomes containing known family members (§2.1.1), the pipeline locates each known element with cmsearch, extracts the flanking coding sequences, and clusters and aligns the encoded proteins. Coding sequences within 1000 nt of each hit were extracted. Proteins were clustered with MMseqs2 (sensitivity 7.5, minimum sequence identity 0.25, coverage $\geq$ 0.2), and clusters with fewer than 5 members were discarded. Italic labels show the data passed between steps. A hidden Markov model is built and calibrated for each cluster (double border, output) and can optionally feed the discovery pipeline as the protein input. Clusters are also analysed for presence/absence and phylogenetic spread across the tree, and searched against InterPro. Hits were called using the Rfam per-family score thresholds and covariance models.

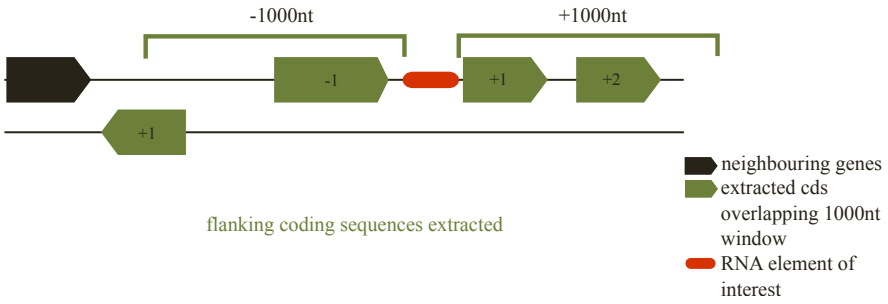


Figure 2.3: Flanking coding sequence extraction. Annotated coding sequences surrounding the RNA element of interest are extracted from genomes within a 1000 nucleotide region. Shown in red is the RNA element of interest identified by cmsearch, in black is a coding sequence outside the 1000 nucleotide extraction region. Indicated in green are the flanking coding sequences extracted for further analysis, labelled by upstream (+) or downstream (-) of the RNA element as well as the order (1 or 2 upstream or downstream). Strandedness of the surrounding coding sequences is also recorded (same or different to the cmsearch hit).

Reference genome annotation files were used to identify the coding sequences flanking the RNA element of interest, using coordinates from the RNA structure search output by a custom script. Protein sequences of these identified coding sequences were then extracted from the genome and used for downstream clustering and alignment.

2.2.2 Clustering and aligning protein sequences

Protein homology was assessed by clustering the extracted protein sequences. Protein sequences were clustered using MMseqs2 v.14.7.0[54]. The clustering was performed with a sensitivity of 7.5, a minimum sequence identity of 0.25, and a minimum coverage of 0.2. The sensitivity value controls the permissiveness of the prefiltering stage, before the alignments are performed; 7.5 is reasonably high and was chosen to ensure that more divergent homologs were captured. The minimum sequence identity and minimum coverage values were chosen similarly to allow inclusion of more divergent sequences. Clusters with fewer than 5 members were discarded to ensure that only clusters with sufficient representation were retained for further analysis.

A Multiple Sequence Alignment (MSA) was built from the cluster protein sequences (MAFFT v.7.525)[55]. This defines positional similarity between the sequences, indicating which amino acids are more likely to appear at each position. This identifies conserved sequence features and provides a basis for building a probabilistic model.

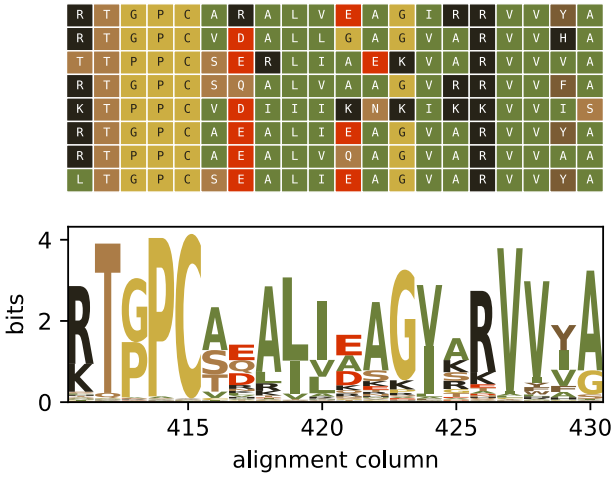


Figure 2.4: Alignment and profile logo of the multiple sequence alignment. A section of a multiple sequence alignment from the most prevalent cluster from the FMN riboswitch flanks (Cluster 204) is visualised with a resulting profile logo. A grid is shown to indicate the position of each amino acid in the alignment (from a selection of 8 cluster members), and the amino acids are coloured by their chemical properties. Below, a profile logo is shown for the alignment section across all sequences, where the height of each letter indicates the relative frequency of that amino acid at that position in the alignment. Amino acid positions are shown on the x axis. Bits on the y axis are a $\log_2$ ratio of the observed frequency of each amino acid at that position against a background frequency.

Probabilistic models, HMM profiles, were built based on the MSA of each cluster. Figure 2.4 shows a section of the MSA of the most prevalent cluster from the FMN riboswitch flanks, and the corresponding profile logo. For each position in the alignment, the profile logo shows the likelihood of each amino acid at that position (quantified in bits). Across a full alignment the profile describes, probabilistically, what a sequence of this cluster looks like. This was done to capture the sequence diversity of each cluster, and to allow for sensitive searches for homologous sequences in the discovery pipeline. HMMER v.3.4 *hmmbuild* and *hmmcalibrate* were used to build and calibrate the HMMs with default parameters.

2.2.3 Analysis of positive control pipeline outputs

The pipeline reports the prevalence of cluster members across the reference genomes. The position of the protein domain relative to the RNA element, the number of clusters identified and their sizes are also reported. Protein domain is defined here reasonably loosly to include a homologous protein sequence which makes up part of multiple proteins.

Taxonomic spread analyses with interactive Tree Of Life (iTOL)

Monophyly or polyphyly of clusters was assessed by mapping the presence or absence of cluster members to the GTDB phylogenetic tree. Additionally an annotation file was generated for each cluster, indicating the presence or absence of a cluster member in each genome. This was visualised on the iTOL v.7 online tool[67] to assess the taxonomic spread of each cluster, and identify bacterial clades where the cluster is more abundant for positive control inputs of the discovery pipeline. The iTOL trees for the protein clusters identified from the FMN riboswitch and CRISPR-Direct Repeat 42 are shown in Appendix Figures 5.1 and 5.2 respectively. Figure 2.5 is illustrative of the taxonomic spread of the clusters.

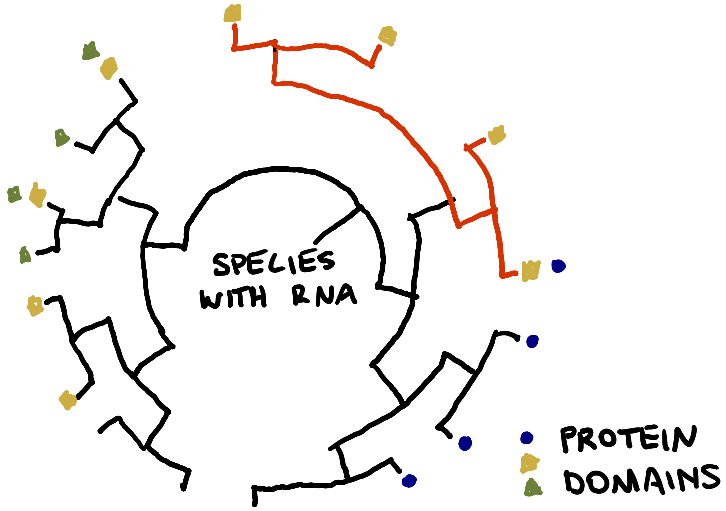


Figure 2.5: Protein cluster member tree. A simplified representation of a phylogenetic tree showing the presence of different protein cluster members across the reference genomes. The tree is annotated with the presence of cluster, where each colour represents a different cluster. The tree structure is based on the GTDB phylogenetic tree narrowed down to representative genomes containing the RNA element of interest (as annotated by Rfam), and the presence of cluster members is based on the positive control pipeline output. In gold is the most prevalent cluster, and indicated in red is a bacterial clade where the gold cluster is more abundant.

Identification with InterPro

Cluster sequences were submitted to the online InterProScan v.6 tool; the results indicated the protein hits and what the protein signal represents[68]. Pfam v.35.0 entries were reported for the highest prevalence cluster, and the InterPro entry was used to identify the protein domain represented by the cluster.

2.3 Computational discovery pipeline

A computational discovery pipeline was developed to identify novel structured RNA elements encoded near homologous protein domains. The pipeline is designed to be flexible and configurable, allowing for the use of different protein domains and reference genome sets. A positive control run was performed with the RibD C-terminal domain, found to be associated with the FMN riboswitch (Section 2.2) over a set of reference *Microbacteriaceae* genomes. A negative control set was also curated and run through the pipeline, to validate the approach and provide a baseline for comparison of the results (Section 2.3.6). Finally the pipeline was run with the BLUF domain over a set of reference *Burkholderia* genomes, to identify novel structured RNA elements associated with this protein domain.

Inputs of the pipeline are a protein domain HMM which serves as an anchor for the search, sourced as described in Section 2.1.2, and a selection of reference genomes as a search space (Section 2.1.1). For validation the protein domain HMM is one recovered from the positive control pipeline (Section 2.2). All inputs from runs of this pipeline used in this study are available in Appendix Table 5.3.

2.3.1 Protein domain search and flanking region extraction

Protein homology was searched using a probabilistic model (HMM) against the protein sequences of the reference genomes (Section 2.1.1) (*hmmsearch*, HMMER v3.4). To narrow the search for structured RNAs, the search was focused on regions flanking the identified protein domains.

The flanking regions are extracted from the reference genomes' genomic sequence based on the coordinates of the protein domain hits and the genomic annotations. Length filtering included only sequences ≥ 20 nucleotides to allow for meaningful homology searches in subsequent clustering steps. Figure 2.7 illustrates the flanking region extraction process, where the anchor coding sequence is identified by *hmmsearch*, and the annotated up and downstream coding sequences of the same strand are used to define the flanking regions.

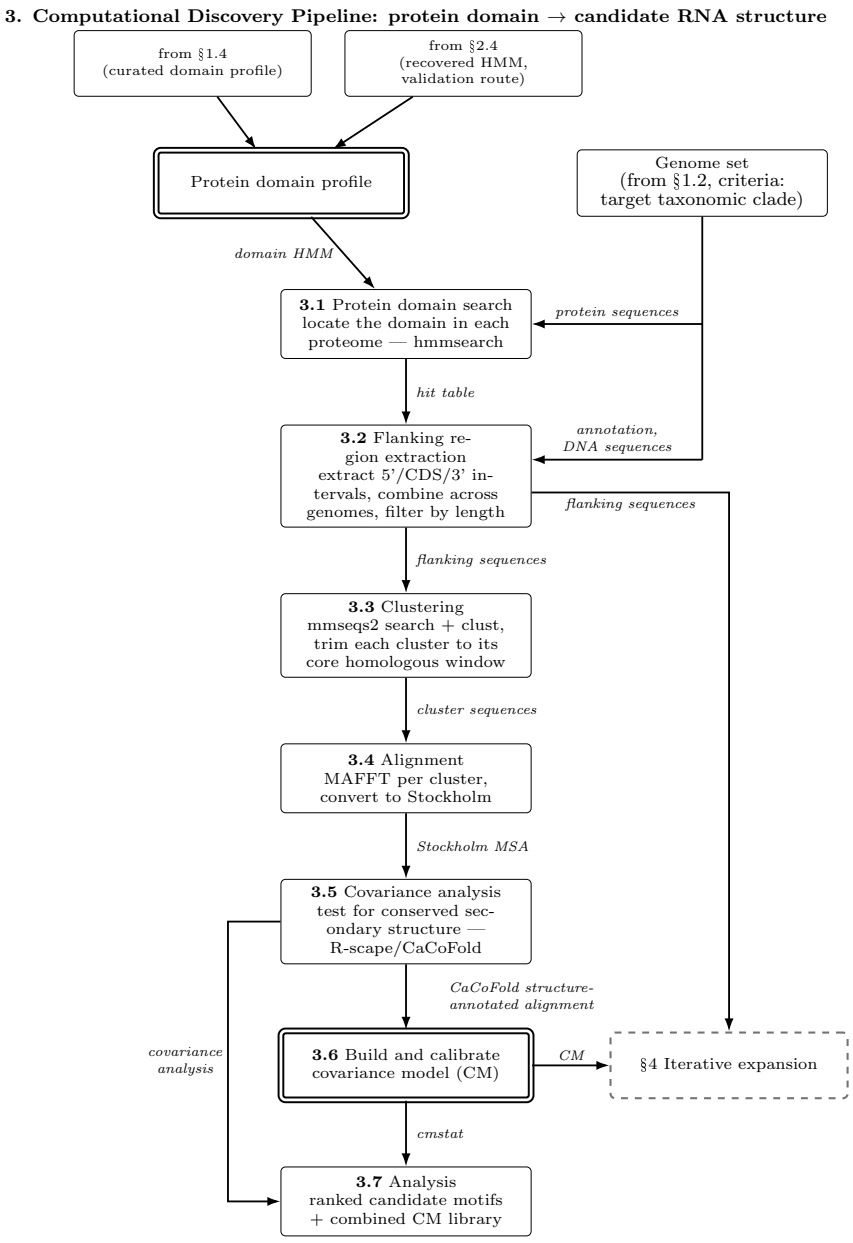


Figure 2.6: Computational discovery pipeline (protein domain → candidate structured RNA). A protein domain profile, either curated (§2.1.2) or recovered from the positive control (§2.2.2), is searched against the proteomes of the target clade (§2.1.1). Flanking regions of each hit are extracted, clustered, aligned and tested for conserved secondary structure, and covariance models are built from the structure-annotated alignments (double border, output). Candidate structured RNAs are ranked in the final report, and the covariance models and flanking sequences pass to iterative expansion (§2.4). Italic labels show the data passed between steps. Filtering thresholds are given in Appendix Table 5.1.

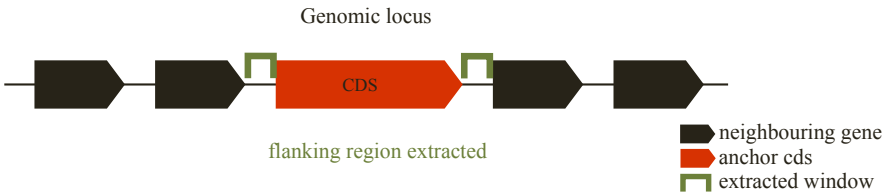


Figure 2.7: Flanking region extraction. Non-coding flanking DNA sequence is extracted from genomes based on hmmer hits in protein sequence and published genome annotations. Shown in red is the anchor coding sequence identified by hmmsearch, in black are annotated up and downstream coding sequences of the same strand. Indicated in green are the flanking regions extracted for further analysis.

2.3.2 Sequence clustering and alignment

Flanking sequences were clustered with a pairwise search based on sequence similarity with MMseqs2 v.16.747c6 *search*[54]. Parameters are defined in Appendix Table 5.1, and were chosen to balance sensitivity and specificity. Ideally clusters would contain sequences that are similar enough to be homologous, but not so similar that they are redundant and reduce the diversity which is required for covariance analysis.

Clustered sequences were then trimmed to the local-alignment window it shares (directly or transitively) with an anchor sequence from the same cluster, using coordinates from a single all-vs-all search pass (MMseqs2 v.16.747c6). Window lengths were filtered to include those of length ≥ 15 and ≤ 1000 nucleotides. A padding of 15 nucleotides was kept on either side of this window, to avoid excluding important sequence information. These trimming and filtering steps were implemented to remove spurious sequence and to focus on the conserved region of the cluster, as some of these sequences were 1000s of nucleotides long.

A multiple sequence alignment was constructed from the trimmed cluster sequences (MAFFT v.7.525)[55]. This identifies conserved sequence features and provides a basis for secondary structure prediction and covariance model building. Alignments also describe how sequences are related to each other, which is important for downstream analyses such as covariance model building and iterative expansion.

2.3.3 Covariance analysis

The CaCoFold algorithm of R-scape v.2.0.4 was used to predict secondary structure and covarying base pairs in the MSA[56]. This approach tests for covariation among pairs of positions in the alignment. It provides a structure-annotated MSA in Stockholm format, which can be used to build a covariance model. Additionally, it provides a statistical test for covariation, giving an E-value for each predicted covarying base pair, statistical power for the covariation test, and an F-measure for the overall align-

ment (the harmonic mean of sensitivity (SEN) and positive predictive value (PPV)). Secondary structure diagrams were generated with R2R v.1.0.6[69].

This covariance analysis step is the key step in the pipeline to identify any structured RNA elements. E-values for covarying base pairs are used to filter out spurious predictions, and the overall F-measure is used to assess the quality of the alignment.

2.3.4 Covariance model construction and calibration

Filtering was done on the CaCoFold structurally annotated alignments to include only those with at least one covarying pair where $E \leq 0.05$. The Infernal v.1.1.5 *cmbuild* tool was used to generate probabilistic models from identified CaCoFold generated structural alignments, using default parameters. Infernal v.1.1.5 *cmcalibrate* assigns E-values to models by fitting exponential tails to the hits of random sequences; this is required for statistical significance in searches using these models[57]. CM with calibration runtimes < 3 hours and requiring < 8 Gigabyte (GB) space on 4 Central Processing Unit (CPU) cores were calibrated by fitting the null tail from the top 5% of the histogram. Models exceding these limits were not calibrated and excluded from reporting. The random sequence set size was also limited to 0.8 Megabyte (MB); this trades off E-value accuracy for quicker run times.

2.3.5 RNA-protein complex prediction

Possible interactions between the BLUF-associated candidate RNAs and BLUF domain-containing proteins were explored with AlphaFold 3[70]. For each candidate, a representative RNA sequence from the candidate alignment and a representative BLUF domain-containing protein sequence from the genome neighbouring the RNA hit were used as input (Appendix Table 5.6). Representative sequences were chosen randomly from the available set. The highest-confidence model from each prediction was used, as determined by the model's predicted confidence score. Confidence in the predicted interaction between the RNA and protein chains was assessed with the interface predicted TM-score (interaction predicted Template Modelling score (ipTM)).

A negative control was run with the same procedure to give reference ipTM values for interpreting the candidate scores. The negative control paired the BLUF protein domain sequence with the FMN riboswitch, which has no documented relationship to the BLUF domain.

2.3.6 Negative control

A negative control was used to assess how the discovery pipeline behaves when anchored on protein domains that are unlikely to be associated with a structured RNA

element. A domain cannot be proven to lack an associated RNA, so this control provides an empirical baseline for the covariation signal expected from the pipeline in the absence of a known RNA element, rather than a true set of negatives. Four membrane-associated protein domains were chosen as negative control anchors, on the basis that structured RNAs are rare across genomes and that membrane-associated domains are unlikely to be involved in RNP systems or cis-regulatory RNAs: the SecY protein translocase, the YgaP-like inner membrane protein, the YadA-like C-terminal outer membrane protein domain, and the bacterial flagellin N-terminal helical region. Domain HMM profiles were retrieved as described in Section 2.1.2.

The search space was the genus *Burkholderia*, chosen because all four domains are present in it, with genomes selected as described in Section 2.1.1. Each domain was screened with the discovery pipeline (Section 2.3). As in all runs of the pipeline, a candidate was retained if its CaCoFold alignment had at least one covarying pair with $E \leq 0.05$. The number of covarying pairs and the R-scape F-score of each retained candidate were recorded and used as a baseline against which the positive control and BLUF candidates were compared.

2.3.7 Checking for known RNA families and protein coding among candidates

Candidate clustered sequences were searched against the Rfam database (version 15.0)[66] to check for known RNA families for the BLUF domain discovery run and the negative control runs. Curated inclusion bitscore thresholds were used to identify significant hits, and the number of hits and their associated families were reported. Additionally the candidate sequences (where the R-scape F-score was > 7) were searched with the NCBI BLASTX tool[71]. Significant hits were identified using an E-value threshold of 1×10^{-5}, and the associated protein families were reported. This search was done to rule out misannotated protein coding sequences and intergenic regions that may have included coding sequences of the other strand.

2.4 Iterative expansion

The idea of this iterative expansion approach is to use the probabilistic models (covariance models) built by the discovery pipeline to search for further homologous sequences in the flanking sequence set. This was done by an iterative approach described in Figure 2.8 using the Infernal v.1.1.5 toolkit[57]. The probabilistic models describe sequence and structural information, and as such can be used to search for likely homologs. These models can be used to align those homologs, and new models can be built from the resulting alignments.

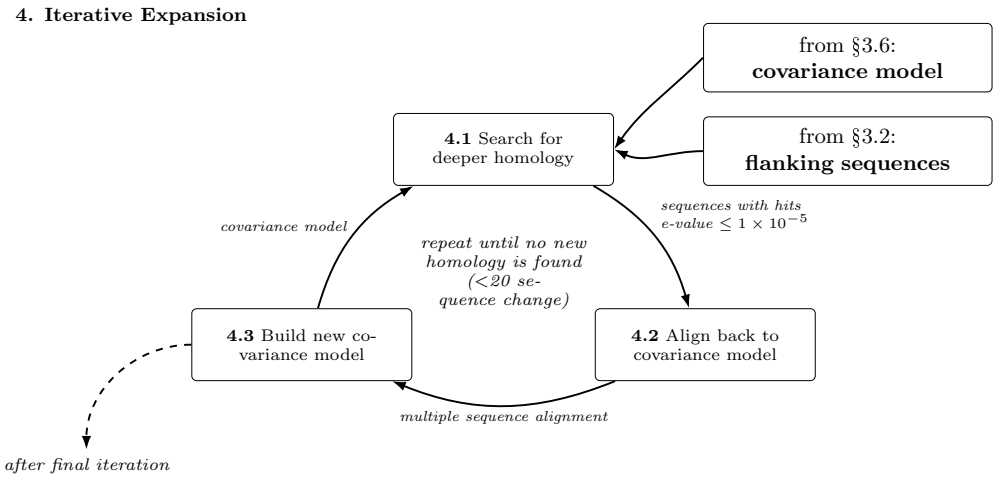


Figure 2.8: Iterative expansion of candidate covariance models. Each covariance model from §2.3.4 is used to search the flanking sequences from §2.3.1 for deeper homology (4.1) with Infernal v1.1.5. Sequences with hits at an E-value $\leq 1 \times 10^{-5}$ are aligned back to the covariance model (4.2), and a new covariance model is built from the resulting multiple sequence alignment (4.3). The cycle repeats until <20 new homologous sequences are found. The dashed arrow marks exit from the loop after the final iteration.

For all uses of this iterative expansion pipeline, a convergence cut-off of 20 new sequences identified in a round was used. This was to avoid long or infinite iterations. When this convergence limit was met a final covariance model and alignment were returned as output.

Search for deeper homology

The probabilistic models (covariance models) built by the discovery pipeline were used to search for further homologous sequences in the flanking sequence set (Infernal v.1.1.5 *cmsearch*). An E-value threshold of 1×10^{-5} was applied to retain significant hits, with otherwise default parameters. Resulting identified sequences were aligned back to the covariance model.

Align back to covariance model

Infernal v.1.1.5 *cmalign* was used to align identified homologous sequences back to the covariance model of interest[57]. This produces a structure-annotated MSA of the identified sequences, which can be used to build a new covariance model, potentially adding new structural and sequence information. These alignments were generated with default parameters as structure-annotated, Stockholm-formatted MSAs.

Build new covariance model

Probabilistic models describing the structure and sequence of the RNA homology identified in the previous steps were built. Infernal v.1.1.5 *cmbuild* and *cmcalibrate* were used to build and calibrate new models from the expanded MSAs[57]. Structure was preserved between rounds with the *cmbuild* `--hand` option, and default *cmcalibrate* parameters.

Output model and alignment analysis

The iterative expansion pipeline outputs are structure-annotated, Stockholm-formatted MSAs and calibrated covariance models. Reported also are the number of rounds of expansion, and the number of sequences identified in each round. Analysis of the output covariance models was done with the covariance model clustering pipeline (Section 2.5).

To assess the accuracy of the discovery pipeline and the iterative expansion, from the FMN riboswitch run a true positive alignment was constructed. This was done by running the Rfam consensus FMN riboswitch covariance model through the iterative expansion pipeline using the identified flanking regions from the discovery pipeline run. The resulting alignment and model answered the question of what was the best possible alignment and model that could be built from the flanking regions. MSAs resulting from positive control candidates run through the iterative expansion pipeline were analysed against the true positive alignment. Coverage identity was calculated using the coordinates of the alignment sequences, as the number of identical nucleotides between alignments divided by the total length of all alignment sequences. Both measures were computed from the sequence identifiers of each Stockholm alignment.

For an alignment Y, let K_Y be the set of sequence keys in Y, and for $k \in K_Y$ let $[s_Y(k), e_Y(k)]$ be the start and end positions of the aligned sequence k (with $s \leq e$).

Let $J\big([s_1, e_1], [s_2, e_2]\big)$ denote the overlap, the intersection over union of the positions they cover, of two intervals $[s_1, e_1]$ and $[s_2, e_2]$. The overlap identity of two alignments A and B is the mean of this overlap over the sequences they share,

$$\mathrm{OI}(A, B) = \frac{1}{|S|} \sum_{k \in S} J\big([s_A(k), e_A(k)],\ [s_B(k), e_B(k)]\big), \tag{2.1}$$

where $S \subseteq K_A \cap K_B$ is the set of shared sequence keys for which both alignments have interval coordinates. OI lies between 0 and 1, with 1 meaning that every shared hit covers exactly the same positions in both alignments; it is reported as a percentage ($100\,\mathrm{OI}$). Sequences present in only one alignment do not contribute, and strand was not used when comparing intervals. An alignment compared with itself has $\mathrm{OI} = 1$,

and two alignments that share sequences but whose hits never overlap have $\mathrm{OI} = 0$.

All analysis was done with custom R v.4.6.1 scripts, available in the supplementary materials (Appendix 5.1).

2.5 Covariance model clustering

An overview of the approaches taken to quantify relatedness between covariance models, probabalistic models built by Infernal tools which describe structural and sequence conservation of RNA elements, is provided in Figure 2.9. These approaches were developed to quantify relatedness between covariance models and to cluster them based on this relatedness. This is described in more detail in the following subsections, with some background on the shared functions of tese approaches.

Sequence scoring

Sequences were scored against each model with the Infernal v.1.1.5 tool *cmsearch*, giving a log-odds bit score for each sequence against each model.

$$S_{\mathrm{bit}}(x) = \log_2 \left[\frac{P(x \mid A)}{P(x \mid \varnothing)} \right] \tag{2.2}$$

where x is a sequence, A is a covariance model, and $\varnothing$ is the null model of random sequence.

Sequence sampling

Sequences were sampled from each probabilistic model (covariance model) with the Infernal v.1.1.5 tool *cmemit*. Sequences are sampled given the model's sequence and structure information, and are representative of the family described by the model. Sequence is generated by the model's underlying HMM, and structure is from the model's underlying CM.

Null sequence pool generation

Null sequences were generated by an implementation of the Dinucleotide Shuffle algorithm first described by Altschul and Erickson[72]. This approach preserves 2-mer composition while destroying any underlying structure or sequence conservation.

2.5.1 Pairwise similarity scoring approaches

Optimal sequence

This approach approximates a shared ancestor, finding a single sequence that can score highly with both models at once, based on the comparison approach by Eggenhofer et al.[59]. The optimal sequence, s^*, was approximated for each pair of models with a

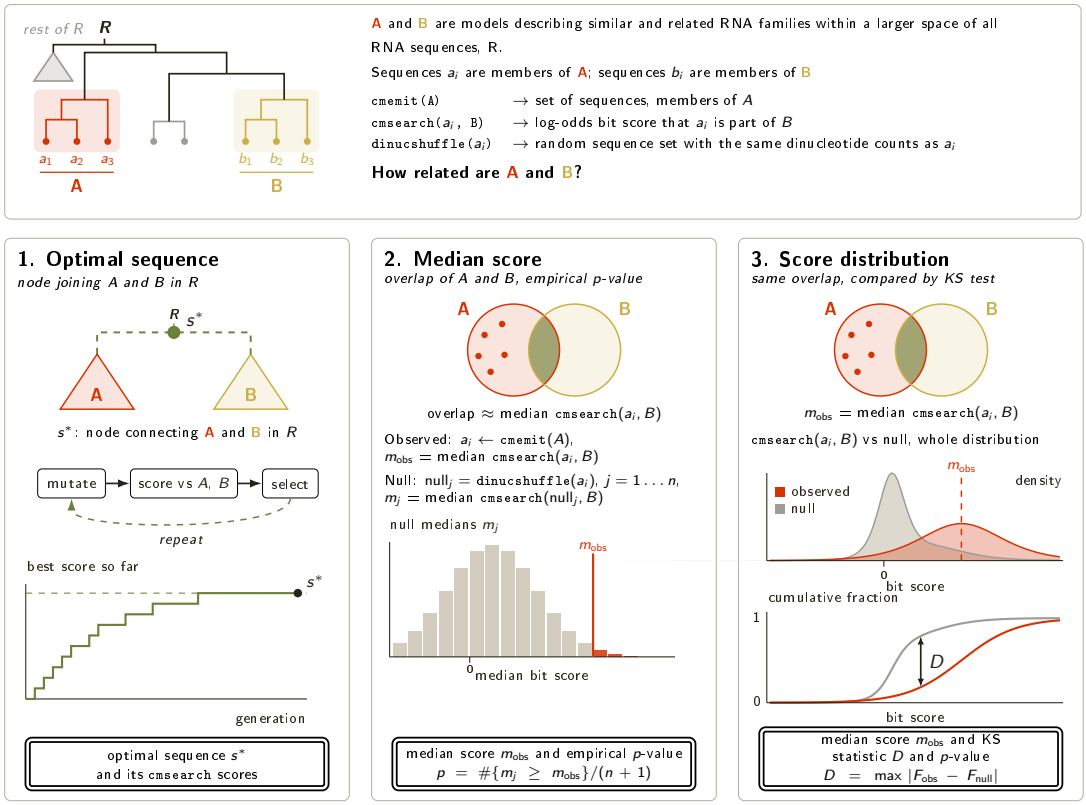


Figure 2.9: Three approaches for quantifying how related two covariance models are. Models A and B describe related RNA families within the space of all RNA sequences, R; sequences a_i are members of A and b_i are members of B. Sequences are sampled from a model with *cmemit*, scored against a model with *cmsearch* (log-odds bit score), and null sequences are generated by dinucleotide shuffling. **(1) Optimal sequence:** a genetic algorithm mutates candidate sequences, scores them against both A and B, and selects the best, repeating over generations to approximate the sequence s^* that joins the two families; the scores of s^* against each model are reported. **(2) Median score:** the median bit score of sequences emitted from A and searched against B (m_{obs}) is compared with the medians of n bootstrap resamples of the pooled scores of dinucleotide-shuffled sequences (m_j), giving an empirical p-value, $p = (1 + \#\{m_j \geq m_{obs}\})/(n+1)$. **(3) Score distribution:** the full distribution of bit scores is compared with the null using a two-sample Kolmogorov–Smirnov test, where the statistic D is the maximum difference between the observed and null cumulative distributions.

genetic algorithm. The initial population was seeded with the consensus sequence of each model and 15 sequences sampled from each model (Section 2.5). In each of 15 generations, every sequence was scored against both models (Section 2.5), and the top 25% by the lower of the two bit scores (against each model) were retained as survivors. The population was then restored to its original size with children, each produced either by crossover of two survivors (probability 0.3) or by mutating one survivor (probability 0.7). Mutation was by point substitution: two positions chosen uniformly at random were each replaced with a nucleotide drawn at random, so sequence length was not changed by mutation. This rate of mutation was chosen to allow for exploration of sequence space at a relatively low rate to retain some of the original sequence information. The search ran for a fixed number of generations (15), and the highest-scoring sequence found was taken as s^*. As the search is stochastic and limited in depth, s^* approximates the optimum and its scores are a lower bound on those of the true optimal sequence.

To make scores comparable between pairs of models, the summed bit score of s^* against both models was normalised by the scores each model gives its own sequences, to give a similarity ratio of models A and B,

$$\mathrm{SR}(A, B) = \frac{s(s^*, A) + s(s^*, B)}{\bar{s}_A + \bar{s}_B} \tag{2.3}$$

where $x_1, \ldots, x_N$ are sequences emitted from model X, so that $\bar{s}_X$ is the mean self score of X. The mean self score is defined as the mean of the scores of the emitted sequences against their own model (the same as the first generation sequence pool). A ratio near 1 indicates that a single sequence scores about as well against both models as typical members score against their own. Implemented as custom Go v.1.27.1 code.

Median score and empirical p-value

This approach asks how well the sequences of one family fit into another family's model, and vice versa. 300 sequences were sampled from each family (Section 2.5) and scored against the other model (Section 2.5).

Let $a_1, \ldots, a_N$ be sequences emitted from model A, and let $s(x, B)$ denote the *cmsearch* log-odds bit score of sequence x against model B.

$$m_{\mathrm{obs}} = \mathrm{median}_{i=1,\ldots,N}\ s(a_i, B) \tag{2.4}$$

To generate null sequences, the sampled sequences were dinucleotide shuffled (Section 2.5). Null populations were bootstrapped from this pool with replacement and median bit scores were used to generate an empirical p-value (all with custom

implementations in Go v.1.27.1).

Each a_i is dinucleotide-shuffled once, $\tilde{a}_i = \mathrm{dinucshuffle}(a_i)$, giving a null pool of the same size as the observed set,

$$\mathcal{P} = \{\, s(\tilde{a}_i, B) : i = 1, \ldots, N \,\}. \tag{2.5}$$

For $j = 1, \ldots, n_{\mathrm{boot}}$, a bootstrap population $\mathcal{P}_j^* = \{x_{j,1}^*, \ldots, x_{j,N}^*\}$ is formed by drawing N scores from $\mathcal{P}$ uniformly with replacement, with median

$$m_j = \mathrm{median}\big(\mathcal{P}_j^*\big), \qquad j = 1, \ldots, n_{\mathrm{boot}}. \tag{2.6}$$

$$p = \frac{1 + \#\{\, j : m_j \geq m_{\mathrm{obs}} \,\}}{n_{\mathrm{boot}} + 1} \tag{2.7}$$

The reported metric of this approach is the sum of m_{obs} for both directions to give a single pair value.

Score distribution

The third approach taken was driven by an observation that although the magnitude of m_{obs} (Equation 2.4) is often small between same clan RNA families (Section 2.5.2), distributional differences between $s(a_i, B)$ and $s(\tilde{a}_i, B)$ can still be seen. The two-sample Kolmogorov-Smirnov (KS) test[73] was applied to the observed scores $s(a_i, B)$ and the null pool $\mathcal{P}$ (Equation 2.5), both of size N. The test statistic, D, is the maximum vertical distance between the empirical cumulative distribution functions of the two sets. It was computed using a custom Go v.1.27.1 implementation and is denoted $D_{A \to B}$ for sequences emitted from A and scored against B. Reported combined score is simply $D_{A \to B} + D_{B \to A}$, used in the preliminary clan benchmark.

2.5.2 Analysis of pairwise similarity scores

To test the practicality and performance of the pairwise similarity scoring approaches, they were applied to a benchmark set of Rfam families curated into clans (Section 2.5.2). All curation and annotation sets and results are available (Appendix 5.1).

Rfam families are curated into clans where they are deemed to be related and share common ancestry[66]. A set of 44 families from various clans was curated from Rfam (version 15.0) to test whether the pairwise similarity approaches could recover clan relationships. All 946 possible pairs of families were scored with each approach.

Separation of within-clan from other pairs was quantified as the area under the receiver operating characteristic curve (or Area Under the Curve (AUC)), calculated from ranks as $\mathrm{AUC} = U/(n_{\mathrm{within}}\, n_{\mathrm{other}})$, where U is the number of (within-clan, other)

pairs of pairs for which the within-clan pair has the higher statistic, counting ties as one half. This is the probability that a randomly chosen within-clan pair outscores a randomly chosen other pair. An AUC of 0.5 indicates no separation and 1 indicates perfect separation. The AUC makes no assumption about the distribution of the statistics and is unaffected by their scale. As pairs sharing a family are not independent, 95% confidence intervals were obtained by bootstrapping over families. 95% confidence intervals for the AUC were calculated by bootstrapping over families, to account for each family appearing in many pairs. In each of 5000 resamples, N families were drawn with replacement from the N families tested, and every pair was retained once for each copy of each of its two families. The AUC was recomputed on the resampled pairs, and the 95% interval is the 2.5th to 97.5th percentile of the resampled values. The same resamples were used for all three approaches, so differences between approaches were assessed by taking the difference in AUC within each resample and reporting the 2.5th to 97.5th percentile of those differences (random seed 1).

Data handling and plotting were done in R v.4.6.1 using the readr and ggplot2 v.4.0.3 packages[74]. Clustering was computed by pairwise distance and plotted with a dendrogram. Plotting and summary functions are available in the repository (Appendix 5.1).

2.6 Ethics, MPI and EPA regulations

This project uses publicly available genomic sequence data and does not involve human participants, animal experimentation, or the construction of genetically modified organisms.

No ethical approval, MPI, or EPA approvals are required at this stage. Should the scope expand to require these approvals, they will be sought.

3. Results

The central focus of this project was to develop a pipeline to identify structured RNA elements and to apply this in the search of candidate light-responsive RNA elements (or "light switches"). This was done by developing a computational discovery pipeline and testing it on a positive control to rediscover the FMN riboswitch, and against a set of negative controls. Further, a set of novel approaches to comparing covariance models was developed and tested on a curated set of Rfam families, and then applied to the positive control candidates to assess their similarity to the FMN riboswitch.

3.1 Positive control search: Can protein homology be rediscovered from known RNPs

The hypothesis on which this work is based is that protein homology can be leveraged to identify RNA homology in RNP or cisregulatory systems, on the basis that they are encoded at neighbouring loci. Approaching the hypothesis from the other direction, the positive control pipeline described in Figure 2.2 searches for protein homology in coding sequences neighbouring known RNA family members. Clustering of neighbouring protein sequences, by pairwise alignment, was used as a proxy for homology.

Table 3.1: Top cluster (by genome prevalence) for each RNA family Top cluster (by genome prevalence) for each RNA family, identified by the positive control pipeline. These are the sourced from the coding regions within 1000nt of the RNA family members, and clustered by protein homology.

RNA family	Cluster	Genomes (prevalence)	Top Pfam InterProScan hit (% of cluster sequences)*	Dominant position (% sequences)**
FMN riboswitch (RF00050)	204	926/2009 (46.1%)	RibD C-terminal domain (100%)	Upstream 1 (97.9%)
SAM riboswitch (RF00162)	831	634/1535 (41.3%)	Cys/Met metabolism PLP-dependent enzyme (100%)	Upstream 1 (97.6%)
CRISPR-DR42 (RF01351)	294	28/65 (43.1%)	CRISPR-associated protein Cas2 (100%)	Downstream 1 (98.9%)
Fluoride riboswitch (RF01734)	761	259/661 (39.2%)	CrcB-like protein, camphor resistance (100%)	Upstream 1 (93.9%)

* Prevalence is the fraction of screened genomes containing the cluster.
** Position is given as strand-relative direction and gene rank (e.g. upstream 1 = RNA element immediately upstream of the cluster gene).

The FMN riboswitch, SAM riboswitch, CRISPR-DR42 and Fluoride riboswitch were all found to be associated with particular conserved protein domains, RibD C-terminal domain, Cys/Met metabolism PLP-dependent enzyme, CRISPR-associated protein Cas2 and CrcB-like protein camphor resistance, respectively (Table 3.1). These protein domains were identified in the highest prevalence cluster of coding sequences neighbouring the RNA family members, and the position of the RNA element relative to the protein-coding gene was consistent with known biology [16;30;75;76].

3.2 Can RNA homology be rediscovered from known RNPs

A selection of riboswitches and the CRISPR repeat element were found to be associated with particular conserved protein domains. With this direction shown it remains to show that RNA homology can be identified anchoring the search with a conserved protein. Any of these could presumably be used for validation of the computational discovery pipeline. The CRISPR-Cas system would be interesting to validate; however, in its current state the discovery pipeline extracts only intergenic regions on the same strand as the protein anchor. The prevalence of the RNP and genome availability (limited by Rfam and Entrez entries) were also deciding factors. As such, the FMN riboswitch and its associated protein domain, the RibD C-terminal domain, were used for positive control validation.

3.2.1 FMN riboswitch as a positive control

The RibD C-terminal domain, with a profile HMM generated in the positive control preparation pipeline (Section 2.2.2), was used as an anchor to investigate if the FMN riboswitch signal could be rediscovered. The bacterial family Microbacteriaceae was used as a search space, informed by the phylogenetic distribution of the riboswitch and protein pairing found in the positive control preparation pipeline (Appendix Figure 5.1). 726 loci were identified in the *Microbacteriaceae* search space (697 genomes), when searched with the curated FMN riboswitch model, E-value $< 1 \times 10^{-5}$. Of these 726 loci, 613 were found in the identified anchor-domain flanking regions. 10 out of 13 candidates of the discovery pipline were able to identify (with at least some overlap) all 613 loci. The outputs of the discovery pipeline were run through the iterative expansion approach described in Section 2.4and the resulting alignments were analysed for covariation with R-scape[56]. These final expanded candidates were compared to the true-positive alignment identified with the Rfam curated FMN riboswitch model. Pipeline thresholds for this run are defined in Appendix Table 5.1. The resulting candidates are shown in Table 3.2, ordered by coordinate overlap identity with the true-positive alignment, along with their covariation analysis F-measures and median summed bit scores against the curated FMN riboswitch model.

Table 3.2: FMN riboswitch rediscovery candidates. A summary of the candidate strucutred elements identified by the discovery and iterative expansion pipelines, and their covariation analysis results. Candidates are compared to a true-positive alignment and consensus model to assess their recovery of the known FMN riboswitch signal.

Candidate	F-score*	% overlap identity†	Summed median score (bits)‡
416	78.95	76.75	36.35
647	73.17	67.99	47.45
35	62.07	75.02	52.55
379	61.18	80.36	61.30
491	60.67	67.95	45.20
40	56.67	74.41	38.15
384	55.70	68.73	38.95
539	53.85	76.99	32.45
531	48.15	78.73	34.35
396	34.04	0.00	3.20
109	32.26	0.00	3.00
442	20.00	0.00	−0.60

* R-scape F-measure for covarying base pairs relative to the secondary structure annotated within each candidate's own alignment.
† Mean coordinate overlap (intersection/union) of the hits shared between the candidate and the true-positive alignment.
‡ Sum of the median `cmsearch` bit scores of synthesised candidate sequences scored against the consensus FMN riboswitch model and synthesised FMN riboswitch sequences scored against the candidate model ($N = 300$ emitted sequences per model).

R-scape F-measure is a harmonic mean of sensitivity and positive predictive value, and is an imperfect measure of the strength of covariation signal in an alignment. Table 3.2 shows 9 candidates with a mean coordinate overlap of $> 67\%$ with the true-positive alignment, and summed median bit scores of > 32. This suggests that the candidates identify largely the same genomic loci as the true-positive alignment, and that their probabilistic models describe much of the same structural and sequence information. The high F-measures of the candidates with high overlap identity and bit scores suggest that the covariance analysis is able to detect the strength of a true conserved signal. Secondary structures were predicted for all candidates and the top two, by highest coordinate overlap identity to the true-positive alignment, are shown in Figure 3.1. The predicted structures of the remaining candidates are shown in Appendix Figure 5.3.

Shown for reference in Figure 3.1 is the consensus structure of the FMN riboswitch, as annotated in Rfam, and the positive control alignment, which was made from the same flanking sequences as the candidates and identified by searching with the Rfam FMN riboswitch model. The CaCoFold predicted structures of the two candidates (379 and 531) illustrate the presence of covarying base pairs in the same regions as the true-positive alignment and consensus structure. As can be seen from the annotated helices, the predicted structures of the candidates are similar to the positive alignment and the Rfam consensus structure, with some variation in the number of helices and their lengths. Candidates 379 and 531 (figures 3.1c and 3.1d) have predicted structures with many of the same helices as the positive control and Rfam consensus structure. Interestingly, candidate 379 (Figure 3.1c) has a predicted structure with a hybrid H2-H6 helix, which is not present in the positive control or Rfam consensus structure. This could represent an alternative conformation of the riboswitch, but it may also be an artefact of the alignment and would require further testing.

It should be noted that this is a single, relatively easy positive control. The FMN riboswitch is highly conserved and well characterised, and the protein domain profile used as the anchor was generated from sequences neighbouring the Rfam-annotated riboswitch. Recovery of this signal is therefore a best-case scenario for the pipeline, rather than a general measure of its sensitivity.

3.2.2 Negative control

Four membrane-associated protein domains were chosen as negative controls on the basis that they were unlikely to be involved in RNP systems (Section 2.3.6). These were the SecY protein translocase, the YgaP-like inner membrane protein, the YadA-like, C-terminal domain outer membrane protein and the Bacterial flagellin N-terminal domain. The search space for all was the genus *Burkholderia*, which was chosen as all

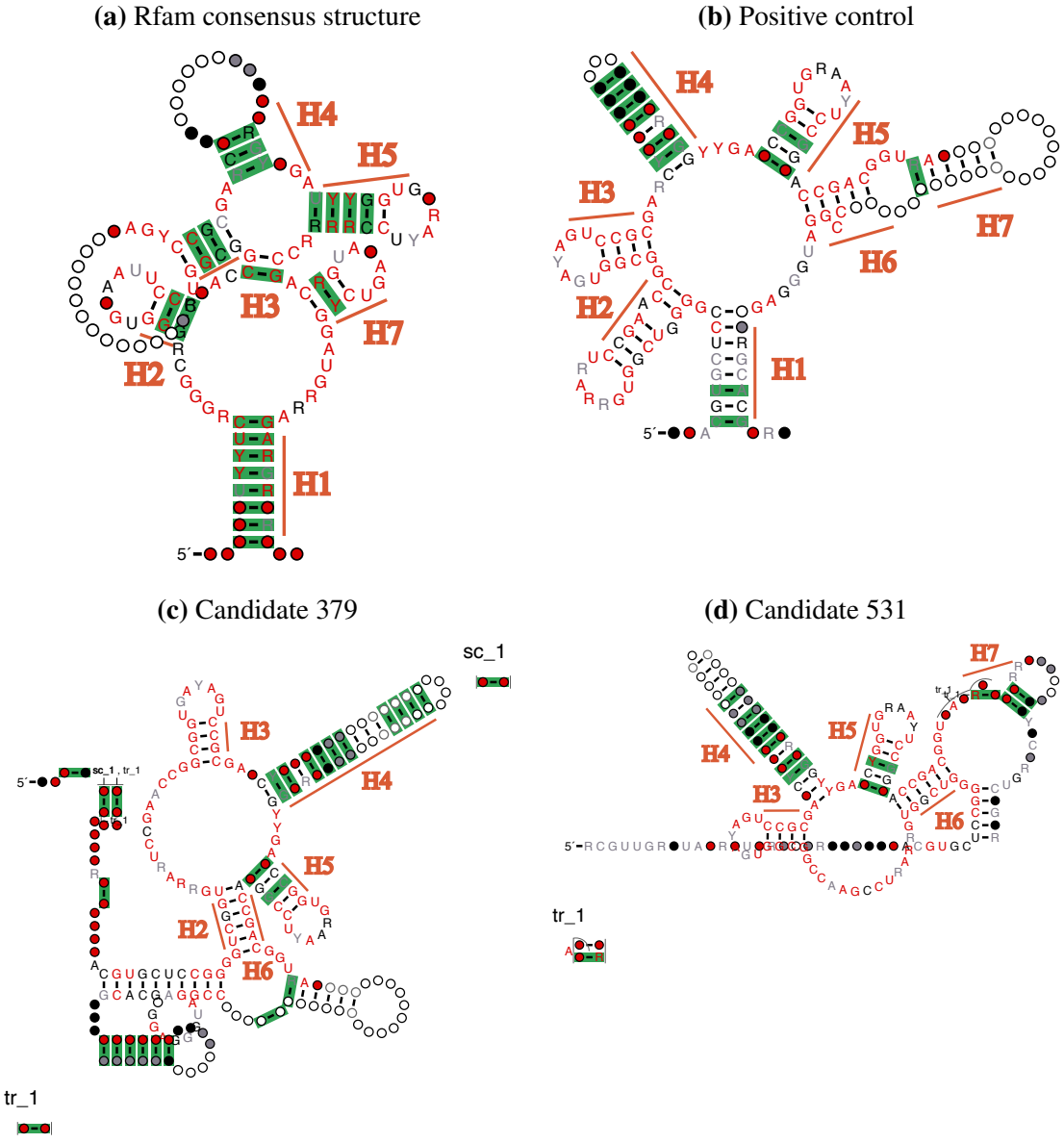


Figure 3.1: Predicted covarying secondary structures of top two FMN riboswitch candidates by highest coordinate overlap identity to the positive control. R-scape CaCoFold structures generated from iteratively expanded alignments, ranked by mean coordinate overlap (intersection/union) of hits shared with the true-positive alignment. Structures are drawn with R2R. Nucleotide letters are shown at positions with a conserved residue and are coloured by conservation (red, $> 97\%$; black, $> 90\%$; grey, $> 75\%$); R and Y are ambiguity codes for purine (A or G) and pyrimidine (C or U). Circles mark positions without a conserved residue and are coloured by occupancy, the fraction of sequences with a nucleotide rather than a gap (clear circles, $> 50\%$). Base pairs with statistically significant covariation ($E \leq 0.05$) are shown in green. Annotated H1-H7 are the seven helices similar across the different structures. **(a)**: the Rfam FMN riboswitch consensus structure, for reference; **(b)**: the true-positive alignment (`fmn_cm_on_micro`), made from the same flanking sequences as the candidates and identified by searching with the Rfam FMN riboswitch model i.e. the best signal that could have been recovered.; **(c)** and **(d)**: the two candidates with highest overlap identity to the true-positive alignment.

four domains are present in this genus.

Table 3.3: Negative control candidates. The SecY protein translocase (IPR002208, PF00538), the YgaP-like inner membrane protein (IPR021309, PF03471) and the bacterial flagellin N-terminal domain (IPR001029, PF00669) were screened in both directions and produced clusters with significant covarying pairs ($E \leq 0.05$). YadA-like membrane anchor domain (IPR005594, PF03895) was also screened in both directions but produced no clusters with a significant covarying pair. Candidates are grouped by domain and ordered by F-measure within each domain.

Label	IPR	Candidate	Direction	Cov. pairs*	F-score
SecY	IPR002208	26	Downstream	1	6.06
SecY	IPR002208	98	Downstream	1	4.44
YgaP-like	IPR021309	90	Downstream	2	5.63
Flagellin	IPR001029	327†	Downstream	11	43.14
Flagellin	IPR001029	545	Downstream	2	8.00
Flagellin	IPR001029	589	Downstream	4	7.48
Flagellin	IPR001029	265	Downstream	1	5.41
Flagellin	IPR001029	16	Upstream	1	4.76
Flagellin	IPR001029	725	Downstream	1	4.35
Flagellin	IPR001029	161	Downstream	1	2.56
Flagellin	IPR001029	272	Downstream	1	2.50
Flagellin	IPR001029	214	Upstream	1	1.20

* Number of covarying base pairs identified by R-scape at $E \leq 0.05$; equal to true positives (TP) since FP = 0 in all cases.
† BLASTX match to pseudaminic acid synthase (see text).

Three of the four negative control domains returned candidates with at least one covarying pair ($E \leq 0.05$): SecY (two candidates), the YgaP-like domain (one candidate) and the flagellin domain (nine candidates). The YadA-like domain returned none.

The highest-scoring negative control candidate was flagellin candidate 327, found in the downstream flanking region, with 11 covarying base pairs and an F-measure of 43.14, which falls within the range of the FMN riboswitch candidates (F-measures of 20.00 to 78.95; Table 3.2). A BLASTX search of this candidate, however, gave a significant match to pseudaminic acid synthase (top hit `WP_478348184.1` from *Mesorhizobium waitakense*; 85.23% identity, 100% query coverage, bit score 449, $E = 6 \times 10^{-157}$; reading frame -1). This indicates that candidate 327 is protein-coding, and that its covariation signal most likely reflects conservation of a protein-coding sequence rather than a structured RNA. The match was in reading frame -1, so the coding sequence lies on the opposite strand to the candidate sequence as extracted. The downstream flanking region therefore contains an opposite-strand protein-coding gene, a case that the same-strand flank extraction of the pipeline does not exclude. Flagellin candidates 545 and 589 (also with F-measures above 7.0) returned no signif-

icant BLASTX hits.

Apart from candidate 327, the eleven negative control candidates had one to four covarying pairs and F-measures of 1.20 to 8.00, much lower than the FMN riboswitch candidates. These provide an empirical baseline for the covariation signal expected from the pipeline when no known structured RNA is present. Candidate 327 shows that protein-coding sequence in the flanking regions can produce a signal comparable to that of a real structured RNA, so the check for coding sequence (Section 2.3.7) is needed before a candidate is interpreted as a structured RNA.

3.3 BLUF domain RNP search

Having tested the discovery pipeline with positive and negative controls, the pipeline was applied to the BLUF domain, which is a light-responsive protein domain. The BLUF domain anchored search was conducted in *Burkholderia*, as this genus contains 701 BLUF domain-containing protein sequences[64].

3.3.1 Identified candidates

The discovery pipeline identified two promising structured RNA candidates, Candidate 130 and Candidate 332, which were found in the downstream flanking sequences of BLUF domain-containing proteins. Large, alignments and those with 0.0 power estimates for covarying pairs were excluded to obtain this set of candidates. These candidates were additionally fed into the iterative expansion pipeline described in Section 2.4, and the resulting alignments were analysed for covariation with R-scape[56].

Table 3.4: Covariation analysis for the two BLUF-associated candidates, before and after iterative expansion of the covariance model. Candidates are identified through the discovery pipeline and then analysed with R-scape for covariation before iterative expansion of the covariance model, over the identified flanking intergenic sequences.

Candidate	Stage	Cov. pairs*	F-score
130	Pre-expansion	4	11.27
130	Post-expansion	0	0.00
332	Pre-expansion	1	2.33
332	Post-expansion	2	5.00

* Number of covarying base pairs identified by R-scape at $E \leq 0.05$; equal to true positives (TP) since FP = 0 in all cases.

Iterative expansion yielded more sequences for both Candidate 130 and 332 (up to 10 and 18 from 6 and 15 respectively); the number of covarying base pairs decreased for Candidate 130 from 4 to none, and increased for Candidate 332 from 1 to 2; similarly, the F-measure increased for Candidate 332 and decreased for Candidate

130 (Table 3.4). This indicates that the original alignment for Candidate 130 better describes a structured element, while the iterative expansion approach improved the alignment for Candidate 332. Therefore, the iterative expansion result was used for Candidate 332, while the original alignment was used for Candidate 130 in subsequent analyses.

The R-scape F-measure is an imperfect measure of the strength of covariation signal in an alignment, shown by the positive and negative controls to have some indicative power. With F-measures of 11.27 and 5.00, respectively, both candidates show only weak evidence of covariation, particularly Candidate 332. Compared with the negative control candidates (Table 3.3), the F-measure of Candidate 130 is higher than all but one (the protein-coding flagellin candidate 327), although only moderately above the next highest (8.00), and the F-measure of Candidate 332 lies within the range of the other negative control candidates (1.20 to 8.00). Another measure of covariance is the number of covarying base pairs. The 4 covarying base pairs of Candidate 130 (original alignment) and 2 of Candidate 332 ($E \leq 0.05$) are within the range of one to four pairs seen for the other negative control candidates, so the number of covarying pairs does not separate them from the negative control baseline. Overall, Candidate 130 shows marginally stronger covariation than the negative controls, whereas Candidate 332 cannot be distinguished from them. Predicted secondary structures are shown in Figure 3.2.

Candidate sequences were also searched against the Rfam database[66] to identify any known RNA families. The annotated family inclusion cutoffs of each model were used to determine if a candidate sequence was a member of a known RNA family. Neither candidate was found to be a member of any known RNA family. Additionally, neither was found to have any significant BLASTX hits, indicating that they are unlikely to be protein-coding sequences.

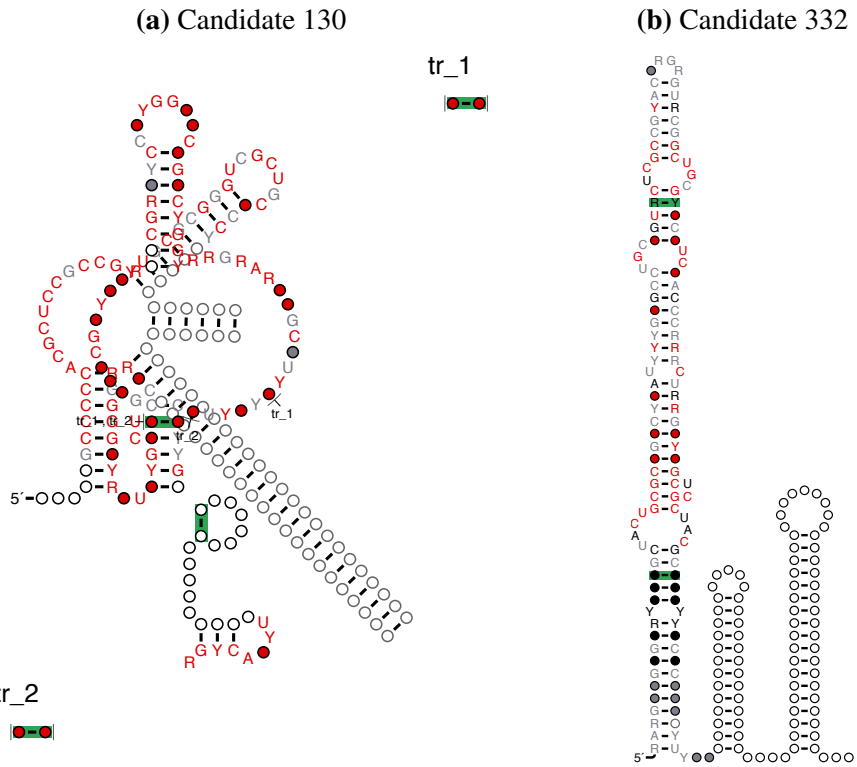


Figure 3.2: Predicted covarying secondary structures of the two BLUF-associated candidates. Predicted covarying secondary structures of the two most promising structured RNA elements identified through the discovery pipeline. Structures are generated by R-scape CaCoFold. Structures are drawn with R2R. Nucleotide letters are shown at positions with a conserved residue and are coloured by conservation (red, $> 97\%$; black, $> 90\%$; grey, $> 75\%$); R and Y are ambiguity codes for purine (A or G) and pyrimidine (C or U). Circles mark positions without a conserved residue and are coloured by occupancy, the fraction of sequences with a nucleotide rather than a gap (clear circles, $> 50\%$). Base pairs with statistically significant covariation ($E \leq 0.05$) are shown in green. Annotated H1-H7 are the seven helices similar across the different structures. **(a)**: shows the predicted secondary structure of candidate 130 as output by the computational discovery pipeline described in the methods, **(b)**: shows that for candidate 332 run through the iterative expansion approach described in the methods.

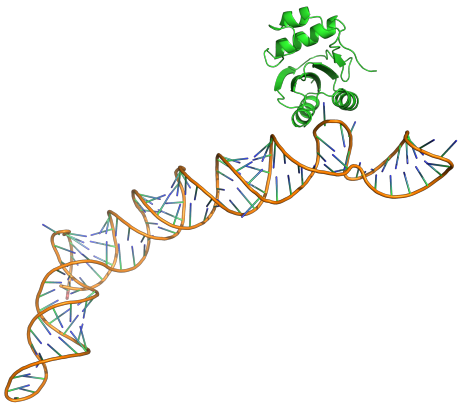
Cancidate 130's predicted secondary structure (Figure 3.2), shows a complex fold with several conserved helices, whereas candidate 332 shows a simpler structure with a couple of conserved helices, while the rest of the structure is less conserved. A wider phylogenetic search space could possibly yield more sequences and stronger covariation signal, making the detection of true conserved structures more robust. Function is not clear from these predicted structures alone, and further experimental work would be required to determine if they are functional structured RNA elements.

3.3.2 RNP complex prediction

3D structure and interaction prediction was done using the highest confidence models from AlphaFold 3[70] (Section 2.3.5) (as determined by the model's predicted confidence

score). Sequences for the following predictions are shown in Appendix Table 5.6.

(a) Candidate 130



(b) Candidate 332

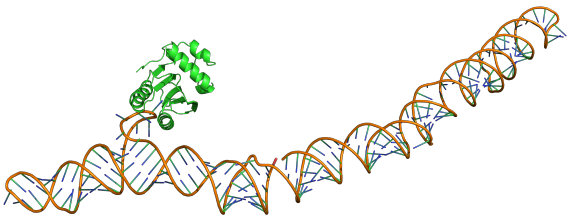


Figure 3.3: Predicted ribonucleoprotein complex structure of BLUF with identified RNA candidates. Predicted structures of representative RNA and protein sequences from the two most promising structured RNA elements identified through the discovery pipeline. Predictions were made with AlphaFold 3 and the highest-confidence model is shown. **(a)** shows the predicted structure and interaction of candidate 130 with the BLUF containing protein it was found near, **(b)** shows that for candidate 332.

RNA binding is predicted around the BLUF FAD binding site identified by Kita et al.[43], although these predictions are low confidence (see below). Appendix Figure 5.5 shows the predicted structure of the BLUF domain with a negative control RNA, which has no documented relationship to the BLUF domain. It predicted an interaction with the protein domain, albeit at a different site to the candidates identified by the discovery pipeline. This shows a clearer interaction with multiple predicted

binding sites.

ipTM scores are an output measure of confidence from the prediction algorithm quantifying the interaction prediction confidence between the two molecules, with a score of 1 indicating high confidence and a score of 0 indicating low confidence. The ipTM scores for the two candidates 130 and 332 are 0.22 and 0.08, respectively. Comparatively, the score for the negative control is 0.1. These scores are all low, and the ipTM score for candidate 332 is below the negative control. Neither prediction therefore provides evidence of a specific interaction between the candidates and the BLUF domain. AlphaFold 3 guide documentation notes that ipTM scores below 0.6 indicate low confidence or failed predictions in the predicted interaction[77].

3.4 Covariance model clustering

Covariance models are the probabilistic models built from multiple sequence alignments to capture the conserved structure and sequences of multiple sequence alignments. Clustering can be done with pairwise scores, ideally one which is reliable and statistically robust. This section describes how three novel scoring approaches were benchmarked against Rfam clans, and how the chosen approach was then applied to cluster the positive control candidates.

Rfam groups families into clans on the basis of shared evolutionary origin, which makes clan membership a useful, if imperfect, ground truth for benchmarking: models in the same clan should recognise each other's sequences, whereas models from different clans should not. Rfam clan membership was therefore used to benchmark the pairwise scoring methods.

3.4.1 Clustering approaches

Three approaches were tested: one which searched for a common ancestor with a genetic algorithm, one which used the median bit scores of sequence populations between the models, and one which looked for distributional differences between the models' sequence population bit score distributions. The median bit score (empirical p value) clan benchmark test took roughly 11 hours over 64 CPU cores, while the optimal sequence and distributional difference approaches took roughly 7 hours over 64 CPU cores. Each was evaluated on its ability to distinguish within-clan pairs from all other pairs (Table 3.5).

All three approaches gave moderate separation of within-clan pairs from other pairs. The similarity ratio and the summed KS statistic performed almost identically (AUC = 0.832 and 0.835), while the summed median bit score against the null performed less well (AUC = 0.767). Because the KS approach provides a per-pair test

Table 3.5: Separation of within-clan from other pairs by each scoring approach. Within-clan ($n = 27$) and other ($n = 919$) pairs from annotated Rfam entries. AUC is the probability that a randomly chosen within-clan pair scores higher than a randomly chosen other pair. Confidence intervals were obtained by bootstrapping over families, to account for each family appearing in many pairs.

Approach	Statistic	AUC	95 % CI
Common ancestor (similarity ratio)	similarity ratio	0.832	[0.716 – 0.934]
Median bit score vs. null	summed medians	0.767	[0.560 – 0.944]
KS distribution vs. null	$D_{A\to B} + D_{B\to A}$	0.835	[0.653 – 0.957]

against a null distribution, which the similarity ratio does not, it was used for all remaining analyses. The distributions of the common ancestor optimal sequece approach and median bit score approach are shown in Appendix Figure 5.6.

3.4.2 Distributional differences to the null

The KS approach compares the bit scores obtained when one model scores the sequences of another against a null distribution of shuffled sequences, in each direction. Figure 3.4 shows this for two pairs.

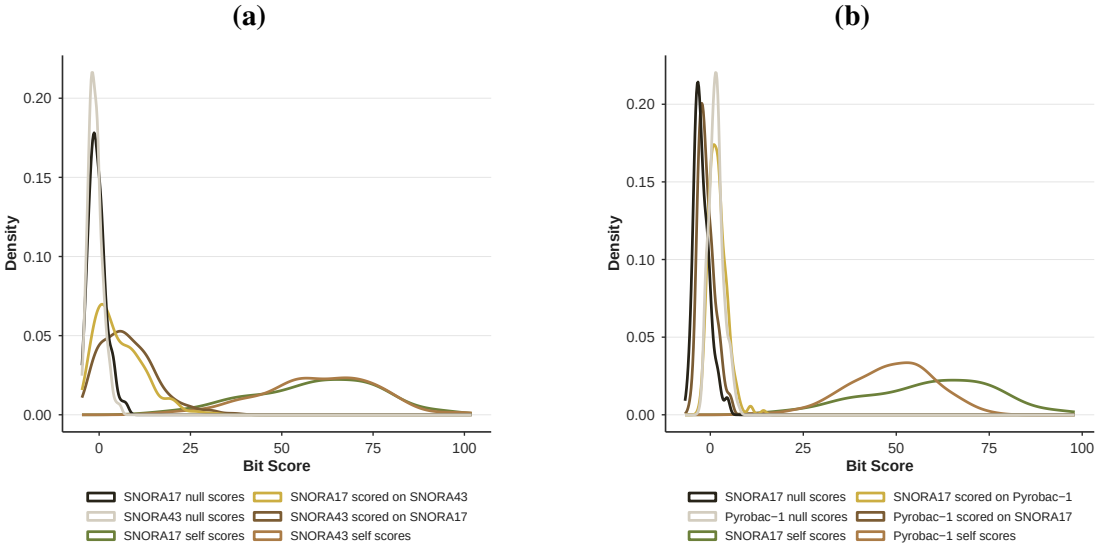


Figure 3.4: Distributional comparison of bitscores for a within-clan and a between-clan model pair. Cross comparison bit score distributions are compared to null sequence distributions. Self bit scores (sequences scored against their model of origin) are shown for comparison. **(a)**: SNORA17 vs. SNORA43, both members of the same clan; the cross-comparison distribution (SNORA43 sequences scored against the SNORA17 model) is compared to null sequences (KS $D = 0.673$, $p = 1.66 \times 10^{-19}$ scored against SNORA43; $D = 0.557$, $p = 3.22 \times 10^{-30}$ scored against SNORA17). **(b)**: SNORA17 vs. Pyrobac-1, from different clans; (KS $D = 0.263$, $p = 5.26 \times 10^{-7}$; $D = 0.112$, $p = 0.112$). ($N = 300$ emitted sequences per model)

The distributions for the within-clan pair (SNORA17 and SNORA43) are wider and shifted to higher bit scores than the null distribution of dinucleotide-shuffled sequences. KS tests are a measure of distributional difference, and make no assumptions about the shape of the distributions[73]. For SNORA17 and SNORA43, which belong to the same clan, the cross-comparison scores were strongly shifted from the null in both directions ($D = 0.673$ and $D = 0.557$; $p = 1.66 \times 10^{-19}$ and 3.22×10^{-30}). For SNORA17 and Pyrobac-1, which belong to different clans, the shift was much weaker in both directions ($D = 0.263$, $p = 5.26 \times 10^{-7}$ and $D = 0.112$, $p = 0.112$). Summing the two directional statistics therefore gives a single, symmetric measure of mutual recognition.

Across all pairs tested, the summed KS statistic was higher for within-clan pairs ($n = 27$) than for other pairs ($n = 919$; Wilcoxon rank-sum $p = 2.76 \times 10^{-9}$, AUC = 0.835; Figure 3.5). Because every model is compared against a single null population it is also considerably cheaper to compute than the bootstrap empirical-p value approach.

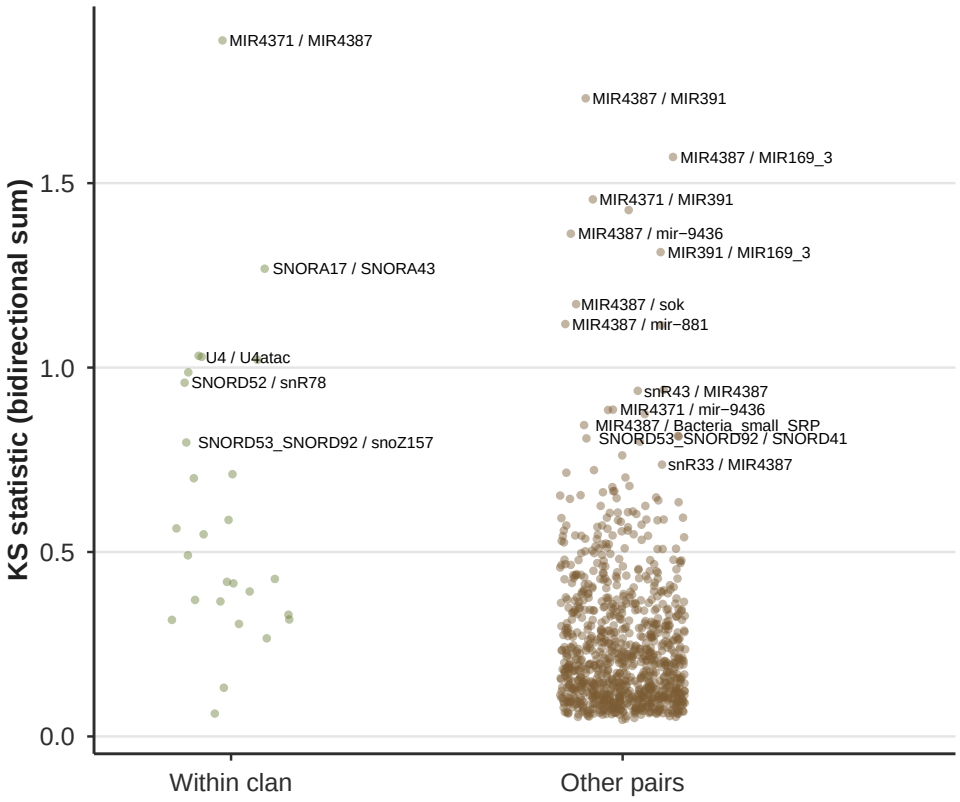


Figure 3.5: Within- vs. between-clan separation using the KS statistic compared against a single null population. The y-axis is the sum of the two two-sample KS statistics, D, of the bit score distribution against the null (both $D_{A \to B}$ and $D_{B \to A}$). A pair is labelled within-clan ($n = 27$) if both families belong to the same Rfam clan, and other ($n = 919$) otherwise, including pairs involving a family with no clan assignment. Top 3% of pairs are labelled with the family names. ($N = 300$ emitted sequences per model)

The summed KS statistic AUC of 0.835 (95% CI: 0.653 – 0.957) shows moderate seperation of within-clan pairs from other pairs of RNA families. There are clearly some family pairs, not in clans, which score highly against each other, noteably the micro RNA families (Figure 3.5).

3.4.3 Clustering positive control candidates

The summed KS statistic was used to cluster the postivie control candidate models together with some reference models, using the distance $2 - (D_{A \to B} + D_{B \to A})$ and average-linkage hierarchical clustering (Figure 3.6). The resulting tree shows candidates clustering into three main groups, one with the consensus FMN riboswitch model clustering with the (`FMN_on_micro`) model and candidates 539, 384, 491, 416, 40, 531, 379, 35 and 647. These close candidates are also the candidates with the highest coordinate overlap identity to the true-positive alignment (Table 3.2). Candidates 442, 109 and 396 do not cluster with the FMN riboswitch model, and are therefore likely to be distinct from the true-positive alignment. 109 and 396 cluster together, which can also be observed from their predicted secondary structures (Appendix Figure 5.4). In downstream analysis these candidates could be combined into a single model, potentially improving the alignment and covariation signal.

The (`FMN_on_micro`) model was constructed with the same iterative expansion approach as the candidates, but using the Rfam FMN riboswitch model to identify sequences in the same flanking regions as the candidates. Through this iterative expansion approach the structure of the input model is preserved. The resulting model is measured to be more similar to the Rfam model than the candidates (Figure 3.6a), although the consensus model was built from a wider range of sequences. Seed alignment analysis revealed that similar seed alignment sequences were used to build the candidates and the (`FMN_on_micro`) model (Table 3.2). The closer clustering of the (`FMN_on_micro`) model with the Rfam model than with the candidates may reflect shared construction, since structure is carried through from the Rfam model during iterative expansion. A control, such as building an FMN model de novo from the same sequences, would distinguish these possibilities.

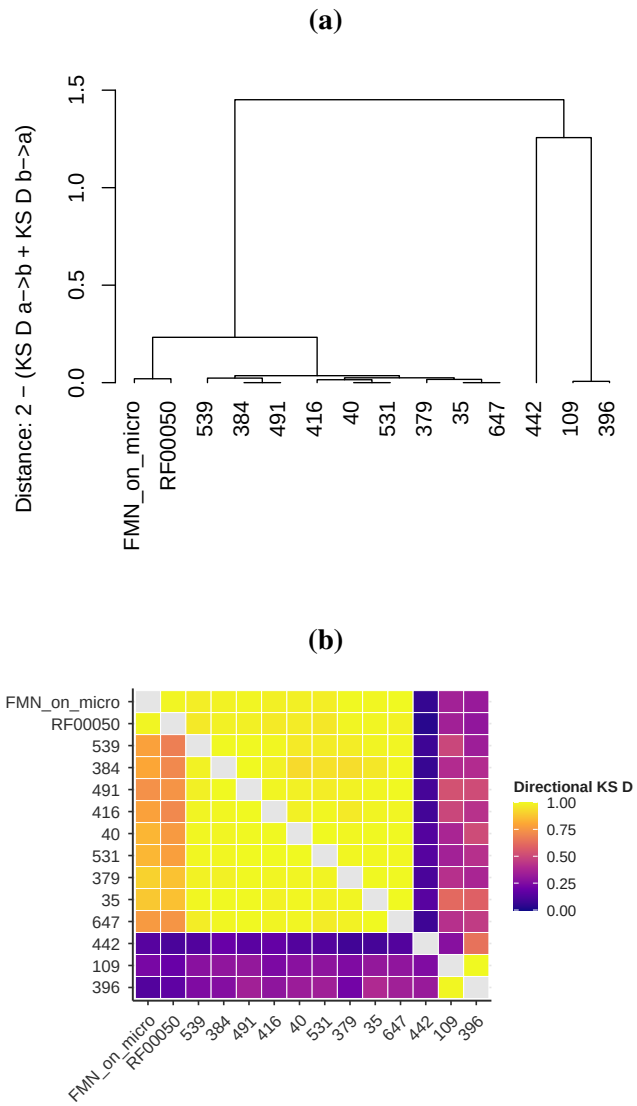


Figure 3.6: Clustering of candidate covariance models by cross-recognition. **(a)** Hierarchical clustering (average linkage) of the candidate models, the Rfam consensus FMN riboswitch model (`RF00050`) and a model built from the consensus model and the same flanking sequences as the candidates (`FMN_on_micro`), using the distance $2-(D_{A\to B}+D_{B\to A})$, where D is the two-sample Kolmogorov–Smirnov statistic against the shuffled null in each direction. **(b)** Pairwise KS statistic, with models ordered as in (a); higher values indicate that the two models recognise each other's sequences more strongly. Candidates are labelled by cluster number. ($N = 300$ emitted sequences per model)

4. Discussion

This project took a protein-first approach to the discovery of structured RNAs, with a focus on RNP complexes. A discovery pipeline, taking a protein domain probabilistic model and bacterial genome space, was developed and then applied to the search for an RNA light switch anchored by the BLUF domain. Further, a novel approach to the clustering of RNA elements was developed.

4.1 Discovery pipeline

The bioinformatic discovery pipeline was the core of this project, and was developed to identify novel structured RNA elements anchored by protein domains. It was tested with a positive control, the FMN-riboswitch-associated RibD C-terminal domain and a negative control set of membrane-associated protein domains. Anchoring the discovery pipeline with the BLUF domain then identified a set of candidate structured RNA elements.

4.1.1 Bioinformatic approach to RNA discovery

The bioinformatic approach to RNA dicovery taken in this project utilised many existing tools and methods, and combined them into a novel pipeline, as well as developing a novel approach to clustering RNA elements. Key tools used in the discovery pipeline were MMseqs2[54] for sequence homology search and R-scape[56] for covariation analysis. Overall the bioinformatic approach was successful in identifying the FMN riboswitch from the RibD C-terminal domain, and in identifying a set of candidate structured RNA elements associated with the BLUF domain.

A variety of other bioinformatic tools were used in this project with varying levels of success and reliability, and many of them were not presented in the results or methods. CMfinder v.0.4.1.18[29] was tested as a candidate method for identifying structured RNA elements, but was found to be unsuitable for this project. It was found to be computationally expensive and to crash on large datasets, such as the FMN riboswitch positive control dataset with 880 sequences, with uninformative error messages. CMfinder has been used successfully by Geissler et al.[42] across 202 genomes, and other stud-

ies[7;8;31], perhaps due to a smaller dataset sizes, or different computational resources. The CMfinder software was also found to be difficult to install and run, with dependencies that were not well documented.

Additionally, CMCompare[59] was tested as a candidate method for clustering covariance models. It was found to no longer be maintained, and the software was not compatible with the current version of Infernal. Web-based CMCompare was also no longer available. Some work went into trying to get the software to run, but it was ultimately unsuccessful.

4.1.2 Positive control

A positive control preparation pipeline was developed to generate a set of positive control sequences for the discovery pipeline. This also provided evidence that modern bioinformatic tools are capable of identifying biologically relevant RNP systems and cis-regulatory elements and protein relationships, from the RNA direction. It was found that the FMN riboswitch is associated with the RibD C-terminal domain, which is a protein domain that is involved in the biosynthesis of riboflavin, and is coded on the rib operon downstream of the riboswitch[76]. The SAM riboswitch was found to be associated with a Cys/Met metabolism PLP-dependent enzyme domain, which matches the *metB* gene (Pfam[65]) found by Tang et al.[75] to be under the control of the SAM riboswitch. Additionally, the Fluoride riboswitch was found to be associated with a CrcB-like protein camphor resistance domain, which is consistent with the known biology of the fluoride riboswitch[30]. These riboswitch examples illustrate the ability to rediscover known riboswitch protein systems begining with a known RNA element, and provided a set of potential positive control pairs for the discovery pipeline.

The CRISPR-DR42 RNA family was used to demonstrate this protein:RNA pair discovery in the context of a known RNP system. The CRISPR-DR42 RNA family was found to be associated with the CRISPR-associated protein Cas2 domain. Cas1 and Cas2 form a complex which binds to CRISPR repeat sequences (like CRISPR-DR42)[16]. This illustrates the ability to rediscover a known RNP system from the RNA direction. Unfortunately the discovery pipeline is currently not set up to extract the intergenic regions of the opposite strand to the identified coding sequence and the CRISPR-DR42 RNA family was found to be located primarily on the opposite DNA strand to the Cas2 coding sequence (Table 3.1).

A single positive control was used: the FMN-riboswitch-associated RibD C-terminal domain, which was chosen for its clear biological association with a protein and its relative conservation across a wide set of bacterial species. A larger set of positive control pairs would better validate the discovery pipeline, and should be considered for future

work.

The discovery pipeline, run using a RibD C-terminal domain anchor, yielded a set of candidate RNA elements with strong similarity to the FMN riboswitch (Figure 3.1, Figure 3.6). This shows, first, that for the FMN riboswitch the intergenic sequences neighbouring a protein domain anchor contain enough information to identify a structured RNA element. Secondly, it suggests that the discovery pipeline can recover this structured RNA element, although sensitivity was not formally quantified. Overall, this provides proof of principle that a protein-first approach to RNA discovery is feasible. This rests on a single, relatively easy positive control, however: the FMN riboswitch is highly conserved and well characterised, and the protein domain profile used as the anchor was generated from sequences neighbouring the Rfam-annotated riboswitch. Recovery of this signal is therefore a best-case scenario, and does not show that the pipeline will detect less conserved or more divergent structured RNAs. Additionally it does not test an RNA protein complex, which is one of the goals of this project.

4.1.3 Negative control

There are many difficulties in the curation of a negative control set for a discovery pipeline stemming from the difficulty of proving a negative. It is impossible to prove that a given protein domain is not associated with a structured RNA element, and so it is difficult to curate a negative control set for the discovery pipeline. The negative control sshould represent the background conservation and random covariance of nucleotide pairs by chance neighbouring conserved protein coding genes. These criteria ruled out randomly generated sequences, or shuffled sequences. Beyond being biologically unrealistic, the covariance analysis tool, R-scape, already tests against a null model of covariation due to phylogeny but independent of structure[56].

As an imperfect solution, a set of membrane-associated protein domains was curated as negative control anchors. The main rationale being that structured RNA elements are rare across genomes[8;42], and so it is unlikely that a random set of protein domains would be associated with structured RNA elements. These were the SecY domain, a protein translocase subunit, YgaP-like domain, a transmembrane protein domain, the YadA-like domain, an outer membrane protein C-terminal domain, and the Flagellin N-terminal domain.

Of the four negative control domains, the YadA-like domain returned no candidates with a significant covarying pair, the SecY and YgaP-like domains returned a few weak candidates (one to two covarying pairs, F-measures of 4.44 to 6.06), and the flagellin domain returned nine candidates. Eight of the flagellin candidates were similarly weak (one to four covarying pairs, F-measures of 1.20 to 8.00), which gives an empirical

baseline for the covariation signal expected when no known structured RNA is present. The ninth, candidate 327, had 11 covarying pairs and an F-measure of 43.14, within the range of the FMN riboswitch candidates. However, a BLASTX search matched this candidate to pseudaminic acid synthase, so its signal most likely reflects conservation of a protein-coding sequence rather than a structured RNA. The match was in reading frame -1, indicating that the coding sequence is on the opposite strand to the flagellin anchor gene, within the flanking region examined by the discovery pipeline.

This outcome is informative rather than a failure of the control. It shows that conserved protein-coding sequence in the flanking regions can generate a strong covariation signal that the covariance analysis alone does not rule out, so the BLASTX check for coding sequence (Section 2.3.7) is a necessary step before a candidate is interpreted as a structured RNA. Alternatively the discovery pipeline could define intergenic regions as the non-coding sequence between two coding sequences, regardless of strand. This was the original implementation but was changed to be more permissive.

The flagellin domain is also a less certain negative control than the other three domains. RNA-based regulation of flagellar gene expression has precedent in the literature, for example small RNAs that regulate the flagellar regulator FlgM in enteric bacteria[78]. A conserved structured RNA near a flagellin gene is therefore not implausible, and the weak flagellin candidates are best treated as background of uncertain status rather than as confirmed false positives. The baseline ranges given above include them; excluding flagellin entirely, the SecY and YgaP-like candidates had F-measures of 4.44 to 6.06 and one to two covarying pairs. Finally, this places the BLUF candidates in context: their F-measures (11.27 and 5.00) and numbers of covarying pairs (4 and 2) are close to, or within, the range of the other negative control candidates, so their covariation signal is not by itself convincing evidence of a structured RNA.

4.1.4 Searching with BLUF

BLUF was chosen as a proof-of-concept search target for its potential to be involved in an RNP 'light switch' complex. Two candidates were identified, with weak covariation signal (F-measures of 11.27 and 5.00, and 4 and 2 covarying pairs). Convincing evidence of covariation would be a higher F-measure and covarying pairs with higher statistical power (these were 0.07 for all candidate 130 pairs, and 0.07 and 0.24 for the candidate 332 pairs). AlphaFold 3 interaction predictions were made for the two candidates with the BLUF domain, and the results were compared to a negative control (BLUF with a random RNA). Convincing evidence of a specific interaction between the candidates and the BLUF domain would be an ipTM score above 0.6, which is the threshold for a confident prediction according to AlphaFold 3 guide documentation[77].

It is also unknown whether these structures, if they exist, are biologically relevant, or what function they might have. Beyond this, they may not interact with the BLUF protein domain at all, and may instead be involved in other biological processes, including acting as cis-regulatory elements. The RNA elements may also be functioning with other neighbouring protein domains, and may simply have been picked up through conservation of coding sequence order and proximity to the BLUF domain. Tamames[79] described the conservation of gene order in prokaryotic genomes, noting high conservation in closely related prokaryotic species. This could be tested with the existing positive control preparation pipeline and a set of representative genomes from a range of prokaryotic species in future work. This approach could indicate if other protein domains are conserved in the neighbouring coding sequences of any other identified homologs of the new candidate RNAs.

4.2 Covariance model clustering approach

4.2.1 Developing a novel approach to covariance model comparison

There is a gap in bioinformatic tools for a fast, working, reliable and statistically defensible similarity scoring for covariance models, the predictive models describing the structure and sequence of conserved RNA elements. The inspiration for covariance model clustering came from a lack of tools to quantitatively compare the positive control candidates to the Rfam annotated FMN riboswitch consensus model. There are clear applications of this approach beyond validating the positive control candidates of this project, including identifying relationships between RNA families for database curations.

Preliminary benchmark testing of the developed approaches was done with a curated set of 44 Rfam families from various clans, and the results of pairwise comparisons were compared to the annotated clan relationships[66]. Clans are curated groups of RNA families from Rfam that are believed to be evolutionarily related, and so are assumed to be more homologous to each other than to other families. This was an imperfect test, as there is no ground truth for the evolutionary relationships of RNA families, and so the annotated clans are only a proxy for homology. For example, many Rfam micro RNA families are not annotated as belonging to a clan, but share similar hairpin structures[80]. This structural similarity was detected by all of the following approaches.

The % coordinate overlap identity reported in the results is a simple measure of how similar the seed sequences of two covariance models are over a small sequence space.

It does not take into account the structure or probabilistic nature of the covariance models. The variable size of the seed aligned sequence set also makes this measure unreliable, as a small seed set will have a higher chance of overlap by chance than a large seed set.

The CMCompare tool[59], although not useable as described earlier, goes about comparing models in an interesting way. It uses the models themselves to calculate a sequence which is likely to be homologous to both models, and then uses the bitscore of that sequence against each model as a measure of similarity. This takes advantage of the evolutionary assumption that a common ancestor of two related RNA elements can be found and that the distance to these is a measure of similarity. Rather than reimplementing CMCompare, a genetic algorithm was developed to estimate a sequence that is likely to be homologous to both models, and then use the bitscore of that sequence against each model (a measure of the probability a sequence belongs to a particular model) as a measure of similarity. This approach was found to reliably rediscover clan relationships in the Rfam database. It however lacks an obvious null model to test against, and so the statistical significance of the similarity score is difficult to interpret.

Covariance models can be used to emit a distribution of sequences that are likely to be homologous to the model. Given two covariance models, the overlap of their emitted sequence distributions can be used to measure their similarity. This overlap can conveniently be measured by the models themselves, which can calculate bitscore likelihoods that the emmitted sequences are homologous to the other model. This overlap thought led to the development of the median score and score distribution pairwise scoring approaches. The two approaches diverge in how they measure the overlap of the two distributions and in how they test against a null model of overlap by chance.

The median score approach measures the overlap of the two distributions by taking the median of the bitscore likelihoods of the emmitted sequences from one model against the other model. Medians were used as opposed to means becuase of the observed skewed distribution of the bitscores (Figure 3.4). This approach fortunately has a reasonably obvious null model: the median of populations of dinucleotide shuffled emitted sequences from one model against the other model. Dinucleotide shuffling preserves the dinucleotide frequencies of the original sequences, and so preserves the overall sequence composition and length[72]. This null therefore acts to test against length and compositional bias of the models, and against the possibility of overlap by chance. It also leads to a simple and intuitive empirical p-value, which is the proportion of shuffled medians that are greater than or equal to the observed median. This approach was found to be effective at clustering Rfam RNA families into their annotated

clans. This approach involved searching many null sequeces against the models which was computationally expensive when compared to the optimal sequence approach.

Also observeable from the distribution of bitscores is that the distributions of related models, although not always particularly right shifted, are often flatter and wider than the distributions of unrelated models and to shiffled sequence distributions (Figure 3.4). This led to the development of the KS D statistic approach, which measures the overlap of the two distributions by comparing their cumulative distribution functions against the null shuffled sequence distributions[73]. This has the advantage of being computationally cheaper than the median score approach, as it does not require searching as many null sequences against the models. This approach was also found to be effective at clustering Rfam RNA families into their annotated clans, and was used to cluster the positive control candidates with the FMN riboswitch.

4.2.2 Applications to iterative combination of covariance models

One application for this clustering approach is to integrate it into a discovery pipeline to identify similar covariance models and merge them to create a new model that captures the shared features of the original models. As could be seen in the positive control candidates, some of the candidates are likely to be homologous to each other, but are not similar enough to be detected by classical homology search methods. If a new model could be created from these candidates, it could be used to identify more distant homologs and improve the sensitivity of RNA discovery pipelines, and power of any covariance analysis. Figure 4.1 illustrates this idea, showing three structured RNA elements that have been identified by local alignment, which are homologous to each other, but are not similar enough to be detected by classical homology search methods.

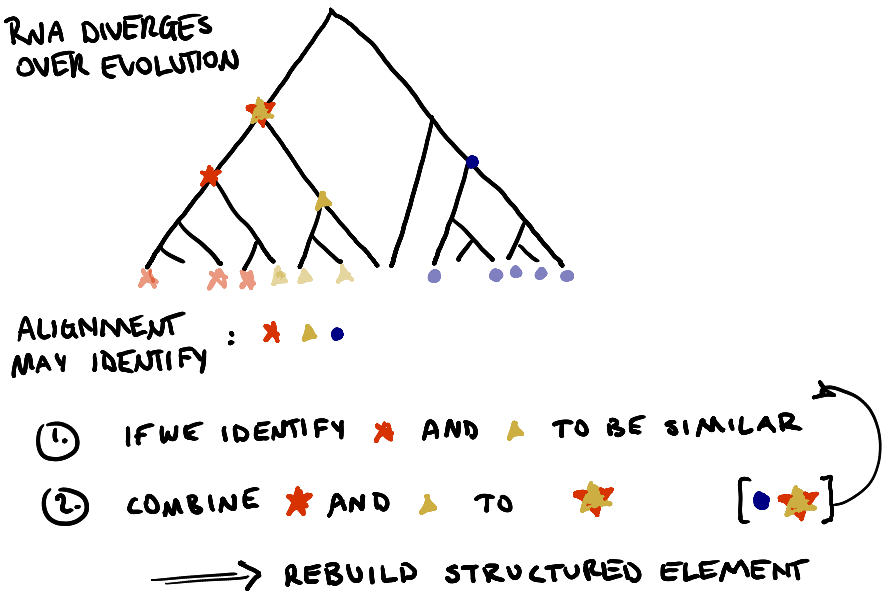


Figure 4.1: Covariance model clustering and merging. Shown in red, gold and blue are structured RNA elements that have been identified by local alignment. Identified elements are homologous to each other, but are not similar enough to be detected by classical homology search methods. Covariance models describing these elements can be clustered based on their similarity, and merged to create a new model that captures the shared features of the original models. This approach can be used to identify more distant homologs and improve the sensitivity of RNA discovery pipelines.

The covariance model combination approach (Figure 4.1) was attempted in this project, but was not successful. Implementation was attempted by aligning synthesised sequences from each model and then building a new model from the resulting alignment. This was either done by aligning the sequences with a multiple sequence alignment tool, or by aligning the models with a covariance model alignment tool. Testing was done with fractured FMN riboswitch models. The Rfam annotated FMN seed sequences (Section 2.1.3) were clustered, aligned, structure predicted and built into covariance models as in the final steps of the computational discovery pipeline (Sections 2.3.2, 2.3.2, 2.3.3 and 2.3.4). Models were then clustered and iteratively merged, and the resulting model was tested against the original FMN riboswitch model. Neither alignment method was successful in producing new models which scored well (with the combined KS D value) against the original FMN riboswitch model. These results were preliminary and not reported. Further work is needed to develop a reliable method for combining covariance models, and to test the resulting models against known structured RNA elements. Also, compared to classical homology search methods, this approach is computationally expensive and time-consuming, and so may not be practical for large-scale RNA discovery pipelines.

4.3 Conclusions

The first aim of this project was to screen for conserved RNA structural elements located near the BLUF domain. This was achieved, and two candidate structured RNA elements were identified. The candidates had weak covariation signal and low-confidence interaction predictions. Further work is needed to validate these candidates, and to determine if they are functional RNAs or involved in light sensing.

The second aim was to develop a protein-first structured RNA discovery pipeline. This was achieved, and the pipeline was tested with a positive control and a negative control set. The positive control FMN riboswitch was successfully recovered. This showed the feasibility of a protein-first approach to RNA discovery. The negative control set provided a baseline for the covariation signal expected when no known structured RNA is present. Most negative control candidates were found to have weak covariation signal. One candidate (flagellin candidate 327) produced a strong false-positive covariation signal, and was found to be a conserved protein-coding sequence rather than a structured RNA. The pipeline is reusable and could be used to screen for conserved RNA structural elements associated with other protein domains in future work.

The third aim was to develop a method for clustering covariance models based on their similarity. This was a later addition to the project, after realising that there was a gap in bioinformatic tools for a fast, working, reliable and statistically defensible similarity scoring. Several approaches were developed and tested, all capable of clustering Rfam RNA families into their annotated clans (Table **??**). Further work is needed to refine this clustering approach and to better validate it.

Overall, the aims of this project were achieved, and a reusable discovery pipeline was developed and tested. There is some evidence for the use of a protein-first approach to RNA discovery, and the discovery pipeline could be used to screen for conserved RNA structural elements associated with other protein domains in future work. Beyond this, there is value in finding structured RNA elements and gaining a better understanding of their biological roles, and in developing bioinformatic tools to support this work.

4.4 Limitations

Several limitations apply to the discovery pipeline and the results presented in this project. The positive control was a single, relatively easy run: the FMN riboswitch is highly conserved and well characterised, and the protein domain profile used as the anchor was generated from sequences neighbouring the Rfam-annotated riboswitch.

Recovery of this signal therefore represents a best-case scenario, and is not a general measure of the sensitivity of the pipeline to less conserved or more divergent RNAs. The negative controls were imperfect, since a domain cannot be proven to lack an associated RNA, only four domains were tested in a single genus, and one of them (flagellin) is itself a less certain negative given reports of RNA regulation of flagellar genes Mika and Hengge[78]. One flagellin candidate scored highly but was protein-coding, so the covariation score alone cannot distinguish a structured RNA from a conserved coding sequence. Overall, controls were limited, partially by the nature of discovery pipelines, but further work should include a larger set of positive and negative controls, and a more diverse set of protein domains and bacterial species.

The BLUF search was restricted to a single genus, *Burkholderia*, and the covariation signal for both BLUF-associated candidates was weak. For each candidate, the original or the iteratively expanded alignment was chosen according to which gave stronger covariation, which adds analytic flexibility. AlphaFold 3 interface confidence scores (ipTM) were low for both candidates and for the positive control, so the predicted complexes provide no evidence of a specific interaction. Finally, no experimental validation was performed, so the candidates should be regarded as hypotheses rather than confirmed functional RNAs.

Screening is also limited to just up and down stream of same strand, as we saw with the CRISPR-DR42 conserved RNA protein associations can go beyond this limited relationship. The discovery pipeline could be extended to mine structured RNAs from more distant intergenic regions, and from the opposite strand to the protein coding gene. Further, the pipeline currently only searches for structured RNAs associated with a single protein domain, but could be extended to search for structured RNAs associated with multiple protein domains, or with a combination of protein domains and other genomic features.

There are also limitations which arrise from the use of unreliable genome annotations. The discovery pipeline relies on the accuracy of genome annotations and proteom translations to identify protein coding genes and their associated intergenic regions. These are known to be error-prone[81], and despite filtering for contamination and completeness, it is likely that some of the genomes used in this project contain annotation errors. It is possible these errors play at least some role in the lack of complete consensus on position relative to homologous protein coding gene for the tested riboswitch cases (Table 3.1).

There will also be limitations in the homology searches, both from the protein domain profile searches and from the clustering and alignment steps. Some of this is unavoidable and a result of the twilight zone of homology, where sequences are too

divergent to be detected by homology search methods[39;40].

Overall, there are many limitations to the discovery pipeline and the results presented in this project, but ultimately is intended to be a quick approach to checking for conserved RNA structural elements located near a protein domain of interest. Once a candidate is identified, further work is needed to validate it and determine if it is a functional RNA or involved in the biological process of interest.

4.5 Future directions

4.5.1 Discovery pipeline extension

There are many ways in which the discovery pipeline could be extended and improved. These include improving the tool selection, filtering steps and tool parameters for the discovery pipeline, and incorporating additional tools for structural clustering and alignment of RNA elements. With a wider range of positive and negative controls, the discovery pipeline could be further optimised for speed and sensitivity. This could introduce some bias in the way current curated RNA elements are identified and lead to overfitting, but would be a useful exercise to improve the discovery pipeline.

The covariance model clustering approach could be integrated into the discovery pipeline to identify similar covariance models and merge them to create a new model that captures the shared features of the original models. This would allow for the identification of more distant homologs and improve the sensitivity of RNA discovery pipelines, and power of any covariance analysis. The alignment and covariance analysis steps would not need to identify deep homology or strong covariation signal, as the clustering and merging of covariance models would allow for the identification of more distant homologs.

There may be other tools that could be used to improve the discovery pipeline, and the pipeline could be further optimised for speed and sensitivity. A preprint from September 2026, Li et al.[82], describes a new approach to the discovery of RNA protein systems, using a large language model to identify non-coding elements coevolving with proteins. Their apprach predicts pairwise interactions between nucleotides and proteins, and has had success with identifying known structured RNA elements and their associated proteins. This approach could be used to identify targets for RNP discovery, or this idea of coevolution could be incorporated into the discovery pipeline to improve the likelihood of finding functional RNP systems. Another tool, RNAalign[83], aligns RNA secondary structures and could be used to improve the clustering of candidate RNA elements. McGimpsey[40] describe an iterative approach to homology search using nhmmer, which they found to lower the twilight zone of RNA homology.

This approach could be used to improve the sensitivity of the nucleotide homology searches in the discovery pipeline, and a similar approach could be taken for the protein domain profile searches. Further works should explore the use of these tools in the discovery pipeline, and compare their performance to the current approach.

Further binary logic could be added to the discovery pipeline to filter candidates based on their genomic context, such as their proximity to other genes or regulatory elements. For example, non coding regions that are located near one light related protein domain or another light related protein domain, or one pathway related protein and a different pathway related protein. This could be used to identify candidates in a more complex biology aware manner.

4.5.2 Further targets for RNP discovery

There are endless possibilities for further targets for RNP discovery, and the discovery pipeline could be applied to a wide range of protein domains and biological processes. Expand beyond BLUF, other light-pathway anchors (e.g. phytochromes, cryptochromes) and other photoreceptor families could be used to anchor searches for further RNP or cis-regulatory candidates. These searches could be extended to other bacterial species, and to other kingdoms of life, including archaea and eukaryotes.

With the success of Durrant et al. [17] in identifying the novel Bridge RNA recombinase system, TIGR from Brück and Randau [27], and the earlier discovery of CRISPR [16], it would be interesting to apply the discovery pipeline to other DNA editing proteins. RNP complex discoveries like this have the potential to be significant biotechnological tools, as the previous examples highlight.

4.5.3 Experimental validation

Once a candidate structured RNA element is identified, and further investigated by computational methods, it should be experimentally validated. Neither of the BLUF-associated candidates identified in this project are currently particularly convincing as structured RNA elements, but if further work found more evidence for them, or if other candidates were identified, they should be experimentally validated. Chemical probing methods, such as SHAPE-seq [33] or DMS-MaPseq [34], could be used to determine the secondary structure of the RNA element. Also, depending on the predicted function of the RNA element, experimental assays could be used to determine its biological role, such as real time quantitative Polymerase Chain Reaction (PCR) and stimulus-response experiments for cis-regulatory elements.

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5. Appendices

5.1 Code and Data Availability

Code is available from:

- https://github.com/dan-432/DarwinsRNAHunt contains the positive control and discovery snakemake pipelines and associated analyses.
- https://github.com/dan-432/cm-iterative-expansion contains the logic for and analysis of the itertive expansion of candidate CMs.
- https://github.com/dan-432/cm-simile contains the logic for and analysis of the pairwise similarity scoring of candidate CMs

See top-level README files for further documentation on file structure and usage instructions.

Data and computed results are available from doi.org/10.5281/zenodo.22927283, all curation and download scripts are available in the code repositories above.

5.2 Supplimentary Tables

Table 5.1: Filtering thresholds used in the computational discovery pipeline. The discovery pipeline was run with these filtering thresholds to identify candidate structured RNA elements associated with the BLUF protein domain. These thresholds were guided by poitive and negative control experiments.

Step	Parameter	Value
3.1 Domain search	hmmsearch E-value	$\leq 10^{-5}$
3.2 Flanking extraction	Minimum length	20 nt
	Overlap into coding sequence	5 nt
3.3 Clustering	mmseqs2 minimum sequence identity	0.4
	Coverage (mode 3)	≥ 0.5
	Minimum cluster size	5
	Trimming padding	15 nt
3.5 Covariance analysis	R-scape E-value (at least one covarying pair)	≤ 0.05
3.6 Calibration	Random sequence length	0.8 megabases

Table 5.2: Rfam families used as positive controls for the RNA-to-protein direction. The positive control pipeline was run with these Rfam families to identify the protein clusters associated with them.

Rfam ID	Description
RF01315	CRISPR RNA direct repeat element
RF00050	FMN riboswitch
RF00162	SAM riboswitch
RF01734	Fluoride riboswitch

Table 5.3: Inputs to the protein-to-RNA discovery pipeline: protein domain (or protein cluster), description, and the GTDB bacterial clade searched. Genomes were filtered at 99.9% completeness and $\leq$1% contamination for every pair.

Domain ID	Description	Bacterial taxon (GTDB node)	Role
IPR005594	SecY domain	*f__Burkholderiaceae_C* (36.0)	Negative control
IPR002208	YadA-like membrane anchor domain	*f__Burkholderiaceae_C* (36.0)	Negative control
IPR021309	YgaP-like transmembrane domain	*f__Burkholderiaceae_C* (36.0)	Negative control
IPR001029	Bacterial flagellin N-terminal helical region	*f__Burkholderiaceae_C* (36.0)	Initially as negative control

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Table 5.3 – continued

Domain ID	Description	Bacterial taxon (GTDB node)	Role
IPR007024	BLUF domain	*f__Burkholderiaceae_C* (36.0)	Test target
cluster_1341	FMN riboswitch control (protein cluster)	*f__Microbacteriaceae* (100)	Positive control
CRISPR_Cas2	CRISPR RNA direct repeat element control (Cas2)	*o__Lactobacillales* (92.0)	Positive control

Table 5.4: Top 10 protein clusters by genome spread for each RNA family. Clusters identified in the positive control pipeline runs for the FMN riboswitch, SAM riboswitch, CRISPR-DR42 and Fluoride riboswitch. Clusters are ordered by prevalence (fraction of available genomes containing the cluster).

Cluster	Genomes	Seqs	Prev.	Position	Up.	Cont.	Strand	Interg.	Purity
FMN riboswitch (RF00050): 1599 clusters, 2009 genomes, 1115 singleton-genome clusters									
204†	926	934	0.4609	Up 1	0.913	0.007	1.0	0.979	0.4609
695	578	748	0.2877	Up 1	0.999	0.003	1.0	0.979	0.288
600	409	417	0.2036	Up 1	1.0	0.01	1.0	0.981	0.3223
89	295	295	0.1468	Up 1	1.0	0.01	1.0	0.98	0.147
58	189	189	0.0941	Down 1	0.0	0.0	1.0	0.979	0.1547
340	188	188	0.0936	Up 1	1.0	0.005	1.0	0.973	0.2709
1453	119	119	0.0592	Down 2	0.0	0.0	1.0	0.975	0.1941
100	96	99	0.0478	Down 1	0.111	0.0	0.869	1.0	0.0478
1501	90	93	0.0448	Down 1	0.462	0.0	0.72	0.935	0.0448
113	71	71	0.0353	Up 1	1.0	0.0	0.0	0.958	0.7474
SAM riboswitch (RF00162): 2247 clusters, 1535 genomes, 1553 singleton-genome clusters									
831†	634	873	0.413	Up 1	0.885	0.006	0.982	0.976	0.413
252	448	450	0.2919	Up 1	0.847	0.082	0.98	0.913	0.2919
2042	406	413	0.2645	Up 1	0.961	0.0	0.884	0.99	0.2645
1591	401	412	0.2612	Up 1	0.993	0.005	0.993	0.973	0.2612
1392	361	586	0.2352	Up 1	0.931	0.015	0.945	0.966	0.2352
2115	262	304	0.1707	Up 1	0.947	0.013	0.977	0.961	0.1707
1725	198	198	0.129	Up 1	1.0	0.0	0.0	0.864	0.3984
1834	134	134	0.0873	Up 1	1.0	0.0	0.0	1.0	0.0873
1517	116	138	0.0756	Up 1	0.992	0.058	1.0	0.935	0.0756
833	98	99	0.0638	Down 1	0.0	0.0	1.0	0.96	0.5904
CRISPR-DR42 (RF01351): 394 clusters, 65 genomes, 340 singleton-genome clusters									
294†	28	264	0.4308	Down 1	0.011	0.0	0.0	0.989	0.4308
217	27	192	0.4154	Down 2	0.068	0.0	0.0	0.99	0.4154
134	24	419	0.3692	Up 1	0.577	0.007	0.024	0.981	0.3692
162	7	65	0.1077	Down 1	0.0	0.0	0.015	1.0	0.1077
121	6	238	0.0923	Up 1	0.542	0.0	0.891	1.0	0.1364

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Table 5.4 – continued

Cluster	Genomes	Seqs	Prev.	Position	Up.	Cont.	Strand	Interg.	Purity
161	6	69	0.0923	Up 1	0.565	0.0	0.188	1.0	0.1364
301	5	37	0.0769	Up 1	1.0	0.0	0.0	1.0	0.0769
120	4	34	0.0615	Down 1	0.0	0.0	0.735	1.0	0.0615
136	4	12	0.0615	Down 2	0.25	0.0	0.333	1.0	0.0615
253	4	20	0.0615	Up 1	0.85	0.0	0.7	1.0	0.0615
Fluoride riboswitch (RF01734): 951 clusters, 661 genomes, 779 singleton-genome clusters									
761†	259	345	0.3918	Up 1	0.974	0.014	0.977	0.939	0.39248
453	254	269	0.3843	Up 1	0.992	0.019	0.981	0.944	0.3848
768	73	89	0.1104	Up 2	0.909	0.011	0.91	0.899	0.1106
561	64	65	0.0968	Up 1	1.0	0.062	1.0	0.877	0.2623
136	33	43	0.0499	Up 1	0.814	0.0	0.744	0.977	0.05
11	28	39	0.0424	Up 1	0.895	0.026	0.795	0.949	0.0424
577	21	28	0.0318	Up 1	1.0	0.036	0.643	0.929	0.0318
169	19	20	0.0287	Down 1	0.4	0.0	0.75	0.85	0.0288
526	18	23	0.0272	Up 1	0.739	0.0	0.87	0.913	0.0272
279	17	17	0.0257	Up 1	0.941	0.0	0.059	0.824	0.0258

Genomes/Seqs: number of genomes/sequences in the cluster; Prev.: fraction of available genomes containing the cluster; Position: dominant position relative to the nearest gene (upstream/downstream and gene rank); Up., Cont., Strand, Interg.: fraction of sequences upstream of, contained within, on the same strand as, and fully intergenic relative to that gene; Purity: clade purity; All clusters were polyphyletic. † marks clusters with an InterProScan hit (see Table 3.1).

Table 5.5: Full R-scape covariation results and pairwise overlap summary for FMN riboswitch rediscovery, and for the Rfam-annotated true-positive alignment. Covariation results for the Rfam-annotated true-positive FMN riboswitch alignment, and for each of the 12 candidate alignments generated by the iterative expansion pipeline. The number of covarying base pairs identified by R-scape at $E \leq 0.05$ is reported, along with the number of false positives (FP). Also reported is the percent overlap identity with the true-positive alignment, and with the nearest candidate alignment to each candidate. The nearest candidate is defined as the candidate with the highest percent overlap identity to the candidate in question. Overlap identity is the mean, over the hits shared between two alignments, of the intersection over union of the hit coordinates (as in Table 3.2).

Candidate	Cov. pairs*	FP	% overlap true pos.	% overlap nearest
fmn_cm_on_micro (positive control)	11	0	100.00	80.36 (379)
416	30	0	76.75	94.87 (531)
647	30	0	67.99	96.65 (491)
35	27	0	75.02	92.87 (379)
379	26	0	80.36	92.87 (35)
491	27	0	67.95	96.65 (647)
40	17	0	74.41	92.95 (531)

Continued on next page

Table 5.5 – continued

Candidate	Cov. pairs*	FP	% overlap true pos.	% overlap nearest
384	22	0	68.73	94.06 (647)
539	14	0	76.99	87.36 (384)
531	13	0	78.73	94.87 (416)
396	8	0	0.00	90.79 (109)
109	5	0	0.00	90.79 (396)
442	2	0	0.00	9.71 (396)

* Number of covarying base pairs identified by R-scape at $E \leq 0.05$; equal to true positives (TP) since FP = 0 for all alignments.

Table 5.6: RNA protein interaction 3D prediction input sequences. Input representative sequences for the AlphaFold 3D structure prediction of RNA structural candidate and its associated BLUF protein. Also included is a negative control RNA sequence, which is a known FMN riboswitch sequence that is not associated with the BLUF protein and a positive control RNA sequence, which is a known Bridge RNA sequence that is associated with a transposase protein.

ID	Description	Sequence
cds-APW43854.1|utr3|CP019239.1:3318396-3318523(-)		
	Representative from BLUF downstream candidate 130	
	RNA	ACGUGGCCCCCACGCUCCGCCGCUGCGCGG GUCGCUGCCCCCCGGGGGGGCUAAUUUGGC UUCGCCGCGCUUCGCGAGCCCUUGGGGCGG CCCGGCGGGAAAUUGCCUUUCCUUCCUCCG UCUACUGA
	Protein	MLVRLLYASRAIDSRPDAIEAILTQSRNYNPSC GITGILCYGGGIFLQAIEGGRMAVSELYGHIQK DARHKDVVLLHYEEISERRFGGWTMGQVNMS KINTNILLKYAEKPELDPYSVSGKVSLALLEEL MATASIIGRA
cds-MFN4120999.1|utr3|JBKBKQ010000028.1:10309-10517(-)		
	Representative from candidate cluster 332	

Continued on next page

Table 5.6 – continued

ID	Description	Sequence
	RNA	AAAGCGGGGCGCUUUGCUACUGCGCCAGCC GGUUCCGGGCCUGCUGUGCUCGCCGUACAG GAGUACGGCUGCGCGCAUCGACCCAGACUA GGCUGGCUCGCUACACAAAUCGCCCCACCU UUCUUCGCAGUCAUCCAUCACGGUUCGUAG CCGCAGCCGCCUAGAAGGCGCUGCGGCUUU UCUGUUCAGGCCGUUCUUUUUCUCGCCCG
	Protein	MLVRLLYASRAVDTSPEAIEAILTQSRQHNPGS GITGVLCYGGGVFLQAIEGGRSAVSELYGHIQK DARHKNVELLHYEEISERRFGGWTMGQVNLS KINHSILLKYSEKPELDPYAVSGKVSFALLEEL MATASIIGRA
BNBA01000017.1/83078-82865 — unrelated FMN riboswitch sequence (negative control)		
	RNA	GAACGUCUUCAGGGCGGGGUGAGAUUCCCC ACCGGCGGUAGGUGCGGCGCCCUUGCGGCG AGGCACGAGCCCGCGAGCGCUUUCGAUGGC AUGCGAUCACGCCCCGCCGCGUGGGUCCGU GCCUUGGAAGGUCAGCAGAUCCGGUCAGAU GCCGGAGCCGACGGUCGGCAGGGCCCGGGG CCAUGCCGGAAAGUCCGGAUGAAAGAAGAC GGUA
	BLUF protein	MLVRLLYASRAVDTSPEAIEAILTQSRQHNPGS GITGVLCYGGGVFLQAIEGGRSAVSELYGHIQK DARHKNVELLHYEEISERRFGGWTMGQVNLS KINHSILLKYSEKPELDPYAVSGKVSFALLEEL MATASIIGRA

5.3 Supplimentary Figures

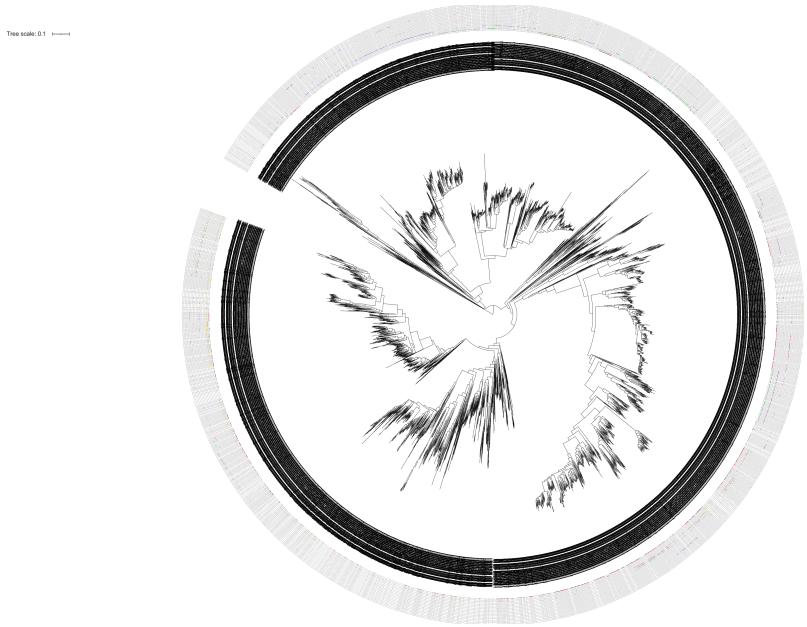


Figure 5.1: Prevalence of homologous protein sequences neighbouring FMN riboswitch. Top 10 clusters by presence per genome are shown around tree, in order of Appendix Table 5.4. Tree is annotated with analysis output of the positive control preperation pipeline output. It shows the GTDB species taxonomy of representative genomes containing the FMN riboswitch (as annotated by Rfam).

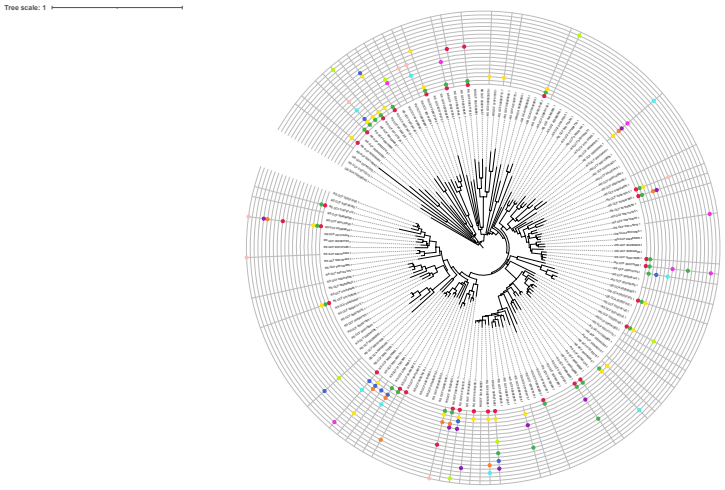
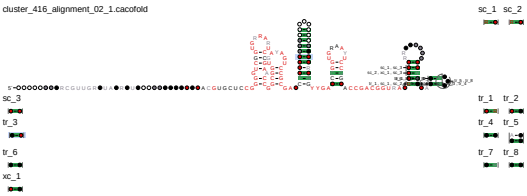
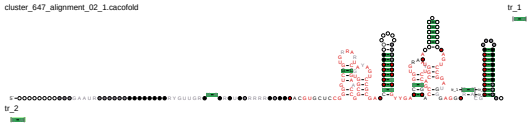


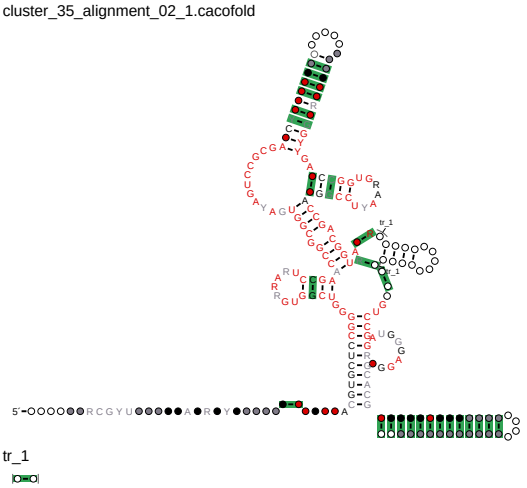
Figure 5.2: Prevalence of homologous protein sequences neighbouring CRISPR-Direct Repeat 42. Top 10 clusters by presence per genome are shown around tree, in order of Appendix Table 5.4. Tree is annotated with analysis output of the positive control preperation pipeline output. It shows the GTDB species taxonomy of representative genomes containing CRISPR-Direct Repeat 42 (as annotated by Rfam).



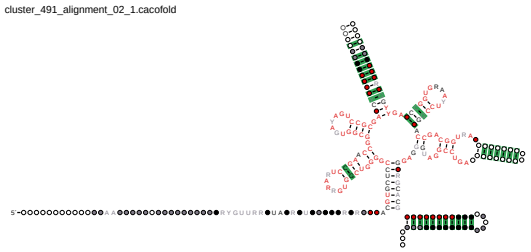
(a) Candidate 416 (F = 78.95)



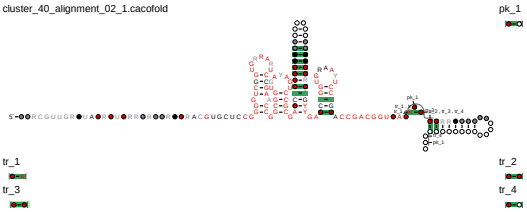
(b) Candidate 647 (F = 73.17)



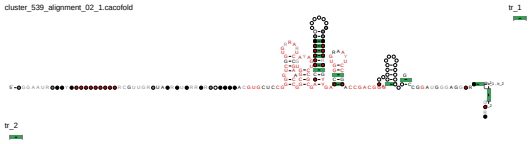
(c) Candidate 35 (F = 62.07)



(d) Candidate 491 (F = 60.67)

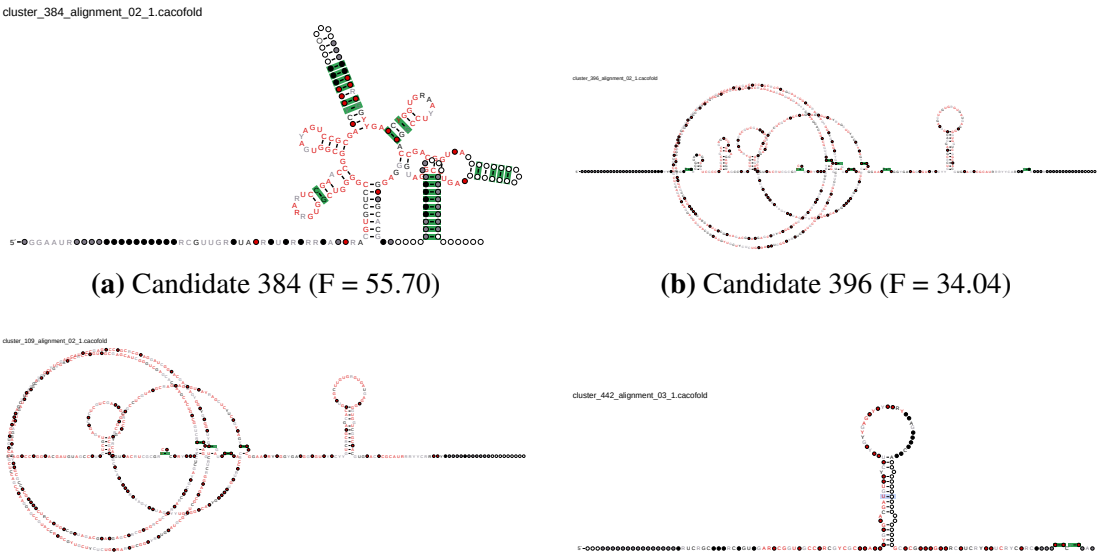


(e) Candidate 40 (F = 56.67)



(f) Candidate 539 (F = 53.85)

Figure 5.3: Predicted covarying secondary structures of remaining FMN riboswitch candidates (part 1 of 2). R-scape CaCoFold structures generated from iteratively expanded alignments, ordered by F-score.



(a) Candidate 384 (F = 55.70)

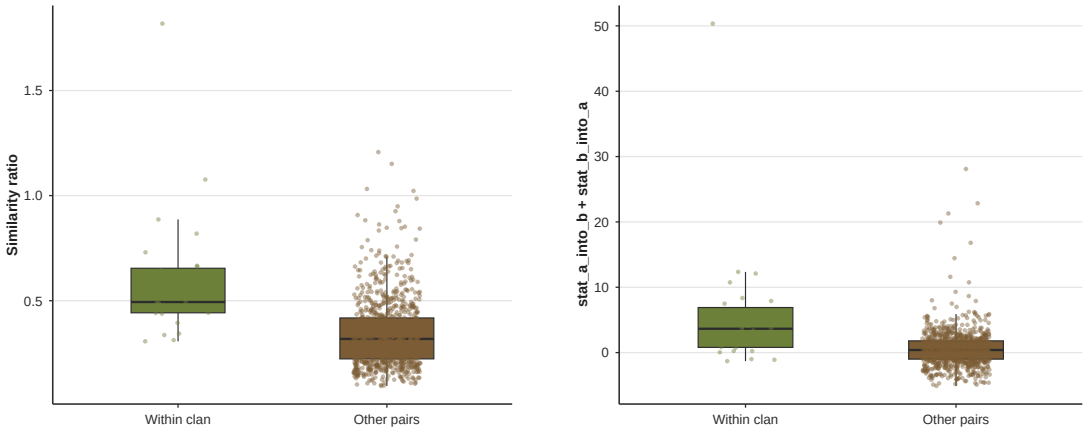
(b) Candidate 396 (F = 34.04)

(c) Candidate 109 (F = 32.26)

(d) Candidate 442 (F = 20.00)

Figure 5.4: Predicted covarying secondary structures of remaining FMN riboswitch candidates (part 2 of 2). R-scape CaCoFold structures generated from iteratively expanded alignments, ordered by F-score. 5.4b and 5.4c correspond to the two low-identity outlier clusters discussed in the results text.

Figure 5.5: Predicted ribonucleoprotein complex structure of BLUF with negative control RNA. Predicted structures of FMN riboswitch and BLUF protein domain sequence. Predictions were made with AlphaFold 3 and the highest-confidence model is shown.



(a) Similarity ratio (minmax)

(b) Bitscore median vs. null

Figure 5.6: Within- vs. between-clan separation for the two initial classification approaches. (a): shows pairwise similarity ratio scores, motivated by common-ancestor theory but lacking a statistical null. **(b)**: shows median bitscore compared against a dinucleotide-shuffled bootstrap null distribution. Within-clan (n = 27) and other (n = 919) pairs are drawn from annotated Rfam entries.